MENTAL HEALTH TEXT CLASSIFICATION VIA FINE-TUNED ROBERTA-LARGE: WITH INTEGRATING LOSS, TRIPLE POOLING, AND CALIBRATION-AWARE TRAINING

Project Report

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CERTIFICATE

This is to certify that the project report entitled "**MENTAL HEALTH TEXT CLASSIFICATION VIA FINE-TUNED ROBERTA-LARGE: WITH INTEGRATING LOSS, TRIPLE POOLING, AND CALIBRATION-AWARE TRAINING**" is a Bonafide record of work carried out By **Borra Khyathi Priyanshu(22481A4216), Kolukula Gnana Sirisha (23485A4206), Gonela Vaishnavi(22481A4233) , Kare Amith Josh (22481A4245)**under the guidance and supervision of **Dr. Y. Adilakshmi**, Professor & Head of the Department, Department of CSE(Artificial Intelligence & Machine Learning) in the partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in CSE(Artificial Intelligence and Machine Learning) of Jawaharlal Nehru Technological University Kakinada, Kakinada during the academic year 2025-26.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
NLP	Natural Language Processing
BERT	Bidirectional Encoder Representations from Transformers
RoBERTa	Robustly Optimized BERT Approach
SVM	Support Vector Machine
TF-IDF	Term Frequency–Inverse Document Frequency
F1-Score	Harmonic Mean of Precision and Recall

ABSTRACT

Mental health disorders represent a growing global concern, and early detection through natural language analysis holds significant promise for scalable screening. This paper presents a fine-tuned RoBERTa-large transformer model for multi-class mental health text classification, capable of categorizing text into seven classes: Normal, Depression, Suicidal, Anxiety, Bipolar, Stress, and Personality Disorder. We propose a classification pipeline that incorporates (I) partial fine-tuning with layer wise learning rate decay, (ii) a triple pooling strategy combining CLS, mean, and max pooling for richer representations, (iii) focal loss with label smoothing to address class imbalance, and (iv) post-training temperature scaling for improved probability calibration. Trained on a dataset of 52,475 original samples augmented with 78,933 back-translated samples (120,913 total training instances), the model achieves a test accuracy of 95.58% and a macro F1-score of 93.48% across all seven classes. Notably, six of the seven classes exceed 92% F1-score, with the majority classes (Normal, Depression, Suicidal) each surpassing 96% F1. We also present an interactive Stream lit-based web application for real-time inference. Experimental results demonstrate that the combination of partial fine-tuning, triple pooling, focal loss, and extensive back-translation augmentation delivers robust performance on this highly imbalanced mental health dataset.
Index Terms—Natural Language Processing, Mental Health, Text Classification, RoBERTa, Transformer, Focal Loss, Transfer Learning, Deep Learning.

KEYWORDS:

Natural Language Processing, Mental Health Analysis, Text Classification, RoBERTa, Transformer Models, Deep Learning, Focal Loss, Class Imbalance, Transfer Learning, Sentiment Analysis, Multi-class Classification, Back Translation Augmentation.

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

Mental health disorders have become one of the most significant global health challenges in recent years, affecting millions of individuals across different age groups, professions, and communities. According to the World Health Organization (WHO), approximately one in eight people worldwide suffer from some form of mental health condition. Disorders such as depression, anxiety, bipolar disorder, stress, and suicidal ideation are among the most common and serious conditions that impact not only an individual's emotional well-being but also their physical health, social relationships, productivity, and overall quality of life. Despite increasing awareness about mental health, access to timely diagnosis and proper treatment remains limited, especially in developing countries and resource-constrained environments. Social stigma, lack of awareness, high costs, and shortage of mental health professionals further prevent individuals from seeking help at the right time, making early detection a critical need.

In the modern digital era, the widespread use of communication platforms such as social media, blogs, online forums, and messaging applications has transformed the way people express their thoughts and emotions. Individuals frequently share their feelings, personal experiences, and mental states through text in the form of posts, comments, and messages. This large volume of unstructured textual data provides a valuable opportunity for analyzing and identifying patterns related to mental health conditions. By examining such textual expressions, it becomes possible to detect early signs of emotional distress, behavioral changes, and psychological issues. This has led to the development of automated systems that can analyze text data and assist in mental health screening and monitoring.

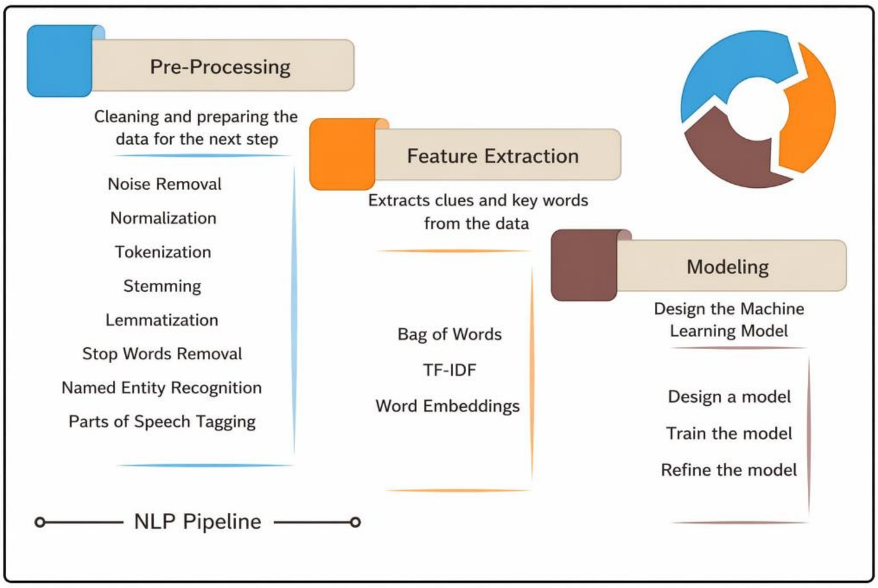

Figure 1.1.1: NLP Pipeline

To address these challenges, this project proposes an advanced mental health text classification system using a fine-tuned RoBERTa-large transformer model. The proposed approach incorporates several innovative techniques to enhance performance and reliability. Partial fine- tuning with layer-wise learning rate decay is used to retain the general knowledge of the pre- trained model while adapting it to the mental health domain. A triple pooling strategy, which combines CLS, mean, and max pooling, is implemented to generate a richer and more comprehensive representation of the input text. This helps in capturing different aspects of the textual data effectively.

In addition, focal loss with label smoothing is applied to handle class imbalance and reduce the impact of dominant classes by focusing more on difficult samples. Temperature scaling is used as a post-training calibration technique to improve the reliability of prediction probabilities and reduce overconfidence. To further enhance the dataset, back-translation data augmentation is performed, where text is translated to another language and back to the original language, increasing diversity and improving model robustness, especially for minority classes.

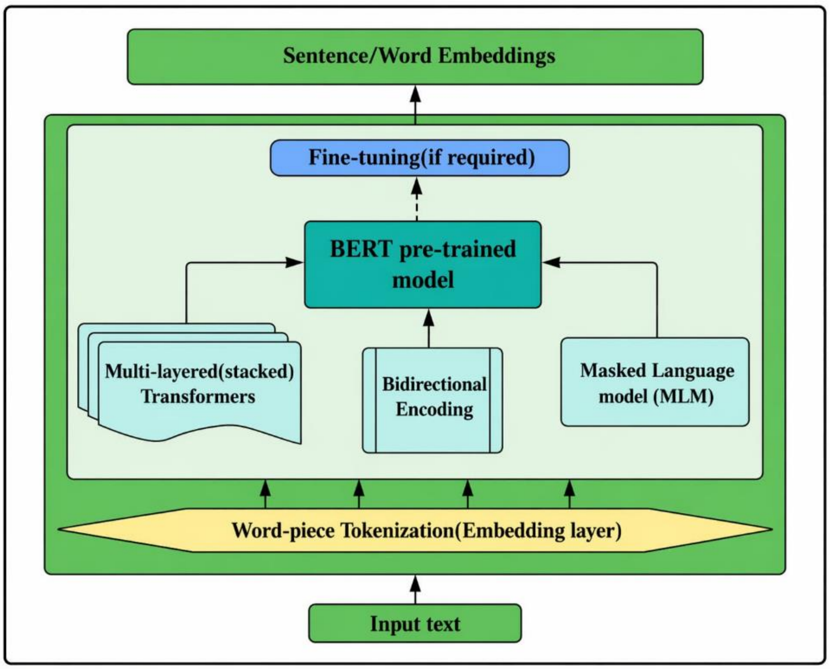

Figure 1.1.2: Word/Sentence Embedding

In addition, focal loss with label smoothing is applied to handle class imbalance and reduce the impact of dominant classes by focusing more on difficult samples. Temperature scaling is used as a post-training calibration technique to improve the reliability of prediction probabilities and reduce overconfidence. To further enhance the dataset, back-translation data augmentation is performed, where text is translated to another language and back to the original language, increasing diversity and improving model robustness, especially for minority classes.

Natural Language Processing (NLP), a sub-discipline of artificial intelligence, provides the methodological toolkit for extracting meaning from unstructured textual data at scale. Advances in NLP over the past decade — from simple bag-of-words models to sophisticated pre-trained transformer architectures — have made it increasingly feasible to analyse subtle linguistic cues, sentiment patterns, and semantic content in ways that correlate with clinical assessments of mental health status. This project is situated within this exciting intersection of AI, NLP, and mental health research, aiming to develop a reliable, scalable, and practically deployable system for automated mental health text classification.

In recent years, the rapid growth of digital communication platforms such as social media, blogs, forums, and messaging applications has transformed the way individuals express their emotions and mental states. People often share their thoughts, experiences, and psychological conditions through textual content, creating a vast amount of unstructured data. This data provides a valuable opportunity for analysing behavioural patterns and detecting early signs of mental health issues. By leveraging such data, automated systems can assist in large-scale mental health screening and monitoring, thereby supporting early intervention and reducing the risk of severe outcomes.

Natural Language Processing (NLP), a key domain of Artificial Intelligence, plays a crucial role in analysing textual data and extracting meaningful insights. NLP techniques enable machines to understand human language by identifying patterns, sentiments, and contextual relationships within text. In the context of mental health analysis, NLP can detect subtle linguistic cues such as negative sentiment, emotional intensity, repetitive thoughts, and self-referential language, which are often associated with psychological conditions. Traditional machine learning approaches relied on handcrafted features such as Bag-of-Words, TF-IDF, and lexical analysis. While these methods provided initial insights, they were limited in capturing deep contextual meaning and complex relationships between words.

However, applying transformer-based models to mental health text classification presents several challenges. One of the primary challenges is class imbalance, where certain categories such as “Normal” and “Depression” dominate the dataset, while others like “Personality Disorder” and “Bipolar” have significantly fewer samples. This imbalance can lead to biased predictions and reduced performance for minority classes. Another major issue is overfitting, especially in large models with millions of parameters, which may memorize training data instead of generalizing well to unseen inputs. Additionally, deep learning models often produce overconfident predictions, which reduces their reliability in real-world applications.

In addition, data augmentation techniques such as back-translation are utilized to increase dataset diversity and improve model robustness, particularly for minority classes. The system is capable of classifying text into seven mental health categories: Normal, Depression, Suicidal, Anxiety, Bipolar, Stress, and Personality Disorder. Furthermore, an interactive web application is developed using Stream lit to provide real-time predictions, making the system practical and user-friendly.

Overall, this project aims to develop a scalable, efficient, and accurate solution for automated mental

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health detection using advanced NLP and deep learning techniques. The system contributes to AI-driven healthcare by enabling early screening, monitoring, and awareness of mental health conditions. However, it is important to note that such systems are intended to assist healthcare professionals and should not replace clinical diagnosis or expert intervention.

1.2 CHALLENGES IN MENTAL HEALTH TEXT CLASSIFICATION

The primary objective of this project is to develop an intelligent and automated system capable of detecting and classifying mental health conditions from textual data using advanced deep learning techniques. In recent years, the increasing availability of user-generated textual data on digital platforms such as social media, forums, and blogs has opened new opportunities for analyzing mental health patterns. This project aims to leverage such data to build a reliable system that can assist in early detection and monitoring of mental health conditions. The system is designed with a strong focus on improving classification accuracy, handling data imbalance, and providing a scalable and practical solution for real-time mental health analysis. By utilizing advanced Natural Language Processing (NLP) techniques and transformer-based architectures, the model is capable of extracting meaningful insights from unstructured text and identifying underlying psychological conditions with high precision.

One of the key objectives of this work is to design a robust multi-class classification model that can accurately categorize textual input into seven distinct mental health classes: Normal, Depression, Suicidal, Anxiety, Bipolar, Stress, and Personality Disorder. These categories are carefully selected to represent a wide spectrum of mental health conditions commonly observed in real-world scenarios. To achieve this, the project utilizes a fine-tuned RoBERTa-large pre-trained model, which is known for its superior ability to capture contextual and semantic relationships within text. Fine-tuning enables the model to adapt to domain-specific language patterns, thereby improving its capability to detect subtle emotional and psychological cues present in user-generated content.

The project also emphasizes effective data preprocessing and preparation, which plays a crucial role in the performance of any NLP-based system. Raw textual data is first cleaned and normalized to remove noise, special characters, and inconsistencies. The processed text is then tokenized using the RoBERTa tokenizer, which efficiently handles subword representations. Padding and truncation techniques are applied to ensure uniform input length across all samples, making the data suitable for batch processing. Furthermore, data augmentation techniques such as back-translation are incorporated to increase dataset diversity and improve model robustness, particularly for underrepresented classes. This helps the model learn a wider range of linguistic variations and improves its ability to perform well in real-world scenarios.

Finally, the project takes into consideration important ethical aspects related to mental health analysis, including data privacy, responsible use of predictions, and potential societal impact. The system is designed to act as a supportive tool for early detection and awareness rather than replacing professional medical diagnosis or treatment. By ensuring ethical and responsible implementation, the project aims to contribute positively to the field of AI-driven healthcare. Overall, the objective is to develop an efficient, scalable, and accurate mental health classification system that leverages advanced NLP and deep learning techniques to address real-world challenges in mental health detection and support.

Furthermore, the project aims to implement and evaluate the system using appropriate performance metrics such as accuracy and F1-score to ensure reliable results. A Streamlit- based web application is also developed to provide real-time text classification, making the system user-friendly and practical for real-world usage.

Finally, the project considers ethical aspects such as data privacy and responsible use of predictions, ensuring that the system acts as a supportive tool for early detection and awareness rather than replacing professional medical advice. Overall, the objective is to develop an efficient, scalable, and accurate mental health classification system using advanced NLP and deep learning techniques.

1.3 MOTIVATION

The primary motivation for this project arises from the growing global burden of mental health disorders and the persistent gap between the demand for mental health services and their availability. According to the World Health Organization, mental health conditions account for a significant proportion of the global disease burden, yet the majority of affected individuals in low- and middle-income countries do not receive adequate treatment. This treatment gap is attributed to a combination of factors including shortage of trained mental health professionals, social stigma, limited healthcare infrastructure, and inadequate awareness about the nature of mental health disorders.

The development of powerful pretrained language models such as BERT and RoBERTa has opened new possibilities for text classification in specialized domains. These models have demonstrated remarkable capability in understanding complex language, capturing subtle semantic nuances, and generalizing across diverse text styles. Applying these models to mental health classification represents a promising approach to improving the accuracy and reliability of automated screening systems beyond what was achievable with earlier machine learning methods. The motivation for this study is therefore driven by both the societal need for scalable mental health support tools and the technological opportunity presented by modern deep learning architectures.

Additionally, the specific technical challenges present in mental health text classification — including significant class imbalance, overlapping symptom expression across conditions, noisy and informal text data, and the need for well-calibrated probability outputs — provide strong motivation for developing and evaluating novel combinations of techniques. The integration of focal loss, triple pooling, back-translation augmentation, and temperature scaling in a single pipeline addresses these challenges in a principled and complementary manner, making this study a meaningful contribution to both the NLP and mental health informatics communities.

However, deploying transformer models for mental health classification presents several challenges:

1. **Class imbalance:**

Mental health datasets are inherently imbalanced, with overrepresentation of common conditions and underrepresentation of disorders like personality disorder. This imbalance can lead to biased predictions where the model performs well on majority classes but poorly on minority classes. As a result, important but less frequent conditions may not be detected accurately, reducing the overall effectiveness of the system. Text data

2. **Overfitting risk:**

Large models with millions of parameters are prone to overfitting on moderately sized datasets. This reduces their ability to generalize well on unseen data and affects real- world performance. Overfitting can cause the model to memorize training data instead of learning meaningful patterns, which limits its practical usability.

3 **Overconfidence:**

Standard SoftMax probabilities are often poorly calibrated, producing overconfident predictions even when the model is incorrect. This can reduce trust in the system, especially in sensitive applications like mental health analysis where reliable predictions are essential.

4 **Subtle distinctions:**

Overlapping symptoms across mental health conditions make multi-class discrimination challenging. Similar language patterns may appear in multiple categories, making accurate classification more complex. For example, expressions of stress and anxiety or depression and suicidal thoughts may share similar words and emotions

In addition to these challenges, noisy and unstructured textual data from online platforms further complicates the classification process. Variations in writing style, use of slang, abbreviations, and incomplete sentences can affect the model's ability to understand context accurately. Therefore, it is important to design robust techniques and preprocessing methods that can effectively handle these issues and improve overall model performance.

1.4 PROPOSED APPROACH

In order to address the challenges associated with mental health text classification, this project proposes a robust, scalable, and high-performance approach based on advanced Natural Language Processing (NLP) and deep learning techniques. The primary goal of the proposed system is to automatically analyse user-generated textual data and accurately classify it into multiple mental health categories. By doing so, the system aims to enable early detection, awareness, and support for individuals experiencing mental health issues. The proposed framework leverages the capabilities of transformer-based architectures, particularly the RoBERTa-large model, which is well known for its ability to capture deep contextual and semantic relationships within textual data.

The overall approach is designed as an end-to-end pipeline that integrates multiple stages, including data preprocessing, tokenization, feature extraction, model training, and prediction. Each stage plays a crucial role in ensuring the effectiveness and reliability of the system. The pipeline is carefully designed to handle real-world challenges such as noisy data, class imbalance, and overlapping linguistic patterns commonly observed in mental health-related text.

The first stage of the approach involves preprocessing of raw textual data. User-generated text from online platforms often contains noise such as special characters, URLs, emojis, abbreviations, and inconsistent formatting. These elements can negatively affect model performance if not handled properly. Therefore, preprocessing techniques such as text cleaning, normalization, removal of irrelevant symbols, and lowercasing are applied to improve data quality. This ensures that the input data is consistent and suitable for further processing.

Following preprocessing, the cleaned text is tokenized using RoBERTa's Byte-Pair Encoding (BPE) tokenizer. Tokenization is a critical step that converts textual input into numerical representations that can be processed by the model. The BPE tokenizer efficiently handles subword units, allowing the system to process rare or unknown words effectively. This is particularly important in mental health datasets, where users may use informal language, slang, or creative expressions. Additionally, padding and truncation are applied to maintain a fixed sequence length, ensuring uniform input dimensions across all samples.

The tokenized input is then passed through the RoBERTa-large encoder, which serves as the core component of the system. The model is partially fine-tuned to balance knowledge retention and domain adaptation. Specifically, the lower layers of the model are frozen to preserve general linguistic knowledge learned during pre-training, while the higher layers are fine-tuned to capture domain-specific patterns related to mental health. This strategy significantly reduces the risk of overfitting and improves the model's ability to generalize to unseen data. Furthermore, layer-wise learning rate decay is applied to ensure stable and efficient training.

Another key component of the proposed approach is the use of focal loss with label smoothing. Mental health datasets are often highly imbalanced, with certain classes having significantly more samples than others. Focal loss addresses this issue by assigning greater importance to difficult and misclassified samples, ensuring that the model learns effectively from minority classes. Label smoothing further enhances generalization by preventing the model from becoming overly confident in its predictions. Together, these techniques improve both accuracy and robustness of the model.

To further improve prediction reliability, temperature scaling is applied as a post-training calibration technique. Deep learning models often produce overconfident predictions, which may not accurately reflect true probabilities. Temperature scaling adjusts the confidence scores without changing the predicted class labels, resulting in better-calibrated probability distributions. This is particularly important in sensitive applications such as mental health analysis, where reliable predictions are essential.

In addition to these techniques, data augmentation is performed using back-translation. This method involves translating text into another language and then back to the original language, generating paraphrased versions of the same content. This increases dataset diversity and helps the model learn different linguistic variations. It is especially beneficial for improving the performance of minority classes and reducing overfitting.

The final stage of the pipeline involves passing the extracted features through a classification head consisting of fully connected layers, followed by a SoftMax function to generate probability distributions over the seven mental health categories. The predicted class is selected based on the highest probability value. The system is designed to provide both the predicted category and its associated confidence score, making the output interpretable and useful for practical applications.

Overall, the proposed approach integrates multiple advanced techniques into a unified framework that effectively addresses the challenges of mental health text classification. The combination of transformer-based modelling, advanced pooling strategies, robust loss functions, and calibration techniques ensures high accuracy, reliability, and scalability. This makes the system suitable for real-world deployment and contributes significantly to the field of AI-driven mental health analysis.

The approach begins with preprocessing of raw text data, where noise, special characters, and irrelevant information are removed to improve data quality. The cleaned text is then tokenized using RoBERTa's Byte-Pair Encoding (BPE) tokenizer, ensuring efficient handling of subword representations. The tokenized input is passed through a partially fine-tuned RoBERTa-large encoder, where lower layers are frozen to retain general linguistic knowledge and higher layers are trained to adapt to the mental health domain. This strategy helps in reducing overfitting while maintaining strong contextual understanding.

Furthermore, the system incorporates data augmentation techniques such as back-translation to increase dataset diversity and improve model robustness, especially for underrepresented classes. The final output is generated using a classification head followed by a softmax layer, which predicts one of the seven mental health categories. Overall, the proposed approach provides an efficient, accurate, and scalable solution for automated mental health analysis, making it suitable for real-world applications and supporting early detection through intelligent text processing.

1.5 SCOPE OF THE STUDY

The The scope of this study encompasses the comprehensive design, development, implementation, and evaluation of an automated mental health text classification system using advanced Natural Language Processing (NLP) and deep learning techniques. The study focuses on analyzing textual data and classifying it into seven predefined mental health categories: Normal, Depression, Suicidal, Anxiety, Bipolar, Stress, and Personality Disorder.

In summary, the scope of this study covers the development of a robust, efficient, and scalable mental health classification system using advanced NLP and deep learning techniques. It aims to bridge the gap between artificial intelligence and mental healthcare by providing a tool that supports early detection and awareness while maintaining ethical and responsible usage.

Scope of this study encompasses the design, development, and evaluation of an automated mental health text classification system based on deep learning and Natural Language Processing techniques. The study focuses specifically on the classification of English-language text into seven predefined mental health categories: Normal, Depression, Suicidal, Anxiety, Bipolar, Stress, and Personality Disorder. The primary dataset used for training and evaluation consists of user-generated text collected from social media platforms and online forums, which represents a realistic source of informal mental health expression in the digital age.

The scope includes the implementation and evaluation of several advanced techniques: partial fine-tuning of the RoBERTa-large transformer model with layer-wise learning rate decay, triple pooling for richer feature extraction, focal loss with label smoothing for handling class imbalance, back-translation data augmentation to expand the training dataset, and post-training temperature scaling for probability calibration. Each of these techniques is evaluated both individually and in combination to assess its contribution to the overall system performance.

1.6 ORGANIZATION OF THE REPORT

The remainder of this report is systematically organized into five chapters, each focusing on a specific aspect of the proposed mental health text classification system. This structured organization ensures clarity, logical progression, and comprehensive coverage of all key components involved in the development and evaluation of the system. Each chapter builds upon the previous one, providing a detailed understanding of the problem, methodology, implementation, and results.

Chapter 2 presents a comprehensive review of existing literature related to mental health analysis using Natural Language Processing (NLP). This chapter explores the evolution of NLP techniques, starting from traditional machine learning approaches to advanced deep learning models. It discusses the development and significance of transformer-based architectures such as BERT and RoBERTa, which have significantly improved the ability of machines to understand human language. In addition, the chapter examines various techniques used in prior research for handling class imbalance, data augmentation, and feature extraction. It also highlights the limitations of existing systems, such as poor generalization, inability to handle noisy data, and lack of reliable probability estimation. These limitations provide the motivation for the development of the proposed approach.

Chapter 3 focuses on the proposed methodology and provides a detailed explanation of the system design and implementation. It begins with the problem formulation, where mental health classification is defined as a supervised multi-class classification task. The chapter then describes the system architecture, including data preprocessing, tokenization, feature extraction, and classification stages.

Special emphasis is given to the fine-tuning strategy of the transformer model, the design of the loss function, and the working of the complete classification pipeline. Each component is explained in a step-by-step manner to provide a clear understanding of how the system processes input text and generates predictions.

Chapter 4 presents the experimental results and discussion, offering a detailed analysis of the model's performance. This includes training performance evaluation, validation metrics, and test set results. The chapter also provides per-class performance analysis to understand how the model performs across different mental health categories. Additional analyses such as ablation studies and error analysis are included to evaluate the contribution of individual components and identify potential weaknesses. Furthermore, the chapter discusses model efficiency, parameter utilization, and computational performance, providing insights into the practicality of the system for real-world deployment.

Chapter 5 concludes the report by summarizing the key findings and contributions of the project. It highlights the effectiveness of the proposed approach in addressing the challenges of mental health text classification. The chapter also discusses the limitations of the current system and suggests possible directions for future research and improvement. Ethical considerations related to the use of AI in mental health applications are also addressed to ensure responsible and safe deployment of the system. The report finally concludes with a detailed bibliography and program outcome mapping, demonstrating the alignment of the project with academic and professional objectives.

Model Design and Fine-Tuning

The system is built upon the RoBERTa-large pre-trained transformer model, which is highly effective in capturing deep semantic and contextual relationships within textual data. Unlike traditional models that rely on handcrafted features, transformer models learn contextual representations directly from data, enabling them to understand complex linguistic patterns. In this project, a partial fine-tuning strategy is adopted to balance knowledge retention and domain adaptation. The lower layers of the model are frozen to preserve general language understanding, while the upper layers are fine-tuned to learn domain-specific patterns related to mental health.

Additionally, layer-wise learning rate decay is applied during training. This technique assigns different learning rates to different layers of the model, with higher layers receiving larger learning rates and lower layers receiving smaller ones. As a result, the model achieves improved generalization performance when evaluated on unseen or real-world textual inputs.

Advanced Feature Representation

To enhance the quality of feature extraction, the proposed system incorporates a triple pooling strategy instead of relying on a single representation. The CLS token captures the overall sentence-level meaning and provides a global representation of the text. Mean pooling computes the average representation across all tokens, capturing the overall semantic context. Max pooling extracts the most important and dominant features by focusing on the highest activation values within the token

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embeddings.

By combining these three representations into a single feature vector, the model obtains a more comprehensive understanding of the input text. This enriched representation significantly improves classification performance, particularly in cases where subtle linguistic variations distinguish different mental health conditions. The use of multiple pooling strategies ensures that no critical information is lost during feature extraction.

Handling Class Imbalance and Prediction Confidence

To effectively handle class imbalance, the system employs focal loss combined with class weighting and label smoothing. Focal loss emphasizes difficult and misclassified samples, ensuring that minority classes receive sufficient attention during training. Label smoothing prevents the model from becoming overly confident in its predictions, thereby improving generalization and reducing overfitting.

This is particularly important in mental health applications, where prediction confidence plays a critical role in decision-making.

Data Augmentation and Robustness

To improve model robustness and handle limited data scenarios, back-translation augmentation is employed. This technique generates paraphrased versions of the original text by translating it into another language and then back to the original language.

The final model achieves high performance with an accuracy of 95.58% and a macro F1-score of 93.48% across seven mental health categories. These results demonstrate the effectiveness of the proposed approach in handling complex and imbalanced datasets. Furthermore, an interactive web application developed using Streamlit enables real-time inference, allowing users to input text and receive instant predictions. This enhances the usability and practical applicability of the system.

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CHAPTER 2

LITERATURE REVIEW

2.1 INTRODUCTION

The field of mental health analysis using Natural Language Processing (NLP) has gained significant attention in recent years, primarily due to the rapid growth of digital communication platforms and the increasing availability of user-generated textual data. Individuals frequently express their emotions, opinions, and psychological states through platforms such as social media, blogs, discussion forums, and online communities. These platforms serve as rich sources of unstructured textual data that can be analyzed to identify patterns associated with mental health conditions. As a result, researchers and practitioners have shown growing interest in developing automated systems that can process, analyze, and interpret such data to support early detection, monitoring, and awareness of mental health issues.

The literature reviewed in this chapter encompasses a broad spectrum of methodologies and technological advancements that have contributed to the evolution of mental health text classification systems. Early approaches in this domain relied heavily on keyword-based and rule-based systems, where predefined lexicons and linguistic rules were used to detect specific mental health indicators. While these methods were simple and interpretable, they lacked scalability and were unable to capture the complexity and variability of human language. Over time, these approaches were replaced by more sophisticated machine learning techniques that leveraged statistical models and feature engineering to improve classification performance.

However, despite their success, RNNs and CNNs had limitations in capturing long-range dependencies and contextual information effectively. This challenge was addressed by the introduction of the transformer architecture, which revolutionized the field of NLP. Transformers utilize self-attention mechanisms to capture relationships between all words in a sentence simultaneously, enabling a more comprehensive understanding of context. This innovation led to the development of large-scale pre-trained language models such as BERT and RoBERTa, which are trained on massive datasets and can be fine-tuned for specific downstream tasks.

In addition to model architecture advancements, the literature also highlights several important challenges and corresponding solutions in mental health text classification. One of the most significant challenges is class imbalance, where certain mental health categories are underrepresented in the dataset. This imbalance can lead to biased models that perform poorly on minority classes. Various strategies have been proposed to address this issue, including class weighting, resampling techniques, and advanced loss functions such as focal loss. These methods aim to improve model performance by ensuring that all classes are adequately represented during training.

Despite these advancements, several limitations remain in existing systems. Many models struggle with generalization across different domains and datasets, especially when dealing with informal or noisy text. Additionally, overlapping symptoms and linguistic similarities between mental health conditions make classification inherently challenging. Ethical concerns, including data privacy, bias, and the risk of misinterpretation, also present significant challenges that must be carefully addressed.

In earlier years, research in this domain mainly focused on traditional machine learning approaches, where manually engineered features were used to classify text into different categories. While these methods provided some level of accuracy, they were limited in understanding deeper contextual meaning and semantic relationships. The complexity of human language, especially in mental health contexts, made it difficult for such systems to achieve high performance.

2.2 NLP FOR MENTAL HEALTH DETECTION

Natural Language Processing (NLP) has emerged as a powerful and effective tool for analyzing textual data in mental health applications. With the increasing use of digital platforms, individuals frequently express their thoughts, emotions, and psychological states through written text. This has created a valuable opportunity for leveraging NLP techniques to automatically detect patterns associated with mental health conditions such as depression, anxiety, stress, bipolar disorder, and suicidal ideation. By analyzing linguistic features, emotional tone, and behavioral patterns present in user-generated text, NLP-based systems can provide meaningful insights into an individual's mental state.

Early research in this domain primarily relied on traditional NLP techniques combined with classical machine learning algorithms. Feature extraction methods such as Bag-of-Words (BoW), Term Frequency–Inverse Document Frequency (TF-IDF), and basic sentiment analysis were widely used to convert textual data into numerical representations. These approaches focused on identifying the frequency of specific words or phrases and detecting general sentiment polarity (positive, negative, or neutral). Machine learning algorithms such as Support Vector Machines (SVM), Naïve Bayes, and Random Forests were then applied to classify the extracted features into different mental health categories. While these methods were relatively simple and computationally efficient, they had significant limitations in capturing contextual meaning and complex linguistic relationships.

Despite significant advancements, challenges still remain in this field. Variability in language usage, presence of noise in user-generated text, and overlapping symptoms across different mental health conditions make classification difficult. Additionally, cultural and linguistic differences can affect the interpretation of text, limiting the generalizability of models across diverse populations. Addressing these challenges requires continuous research and the development of more robust and adaptable NLP techniques.

Overall, NLP has proven to be a valuable tool in the field of mental health detection, enabling automated analysis of textual data with increasing accuracy and reliability. The transition from traditional feature-based methods to advanced deep learning and transformer-based models has significantly enhanced the capability of these systems. These advancements provide a strong foundation for developing intelligent mental health classification systems that can support early detection, monitoring, and awareness in real-world applications.

Recent studies have shown that NLP models can detect subtle indicators of mental health conditions, such as changes in writing style, frequency of negative words, and self-referential language. These findings highlight the importance of advanced NLP techniques in improving mental health detection systems and enabling early intervention.

2.3 PRE-TRAINED LANGUAGE MODELS

The introduction of pre-trained language models has brought a significant transformation in the field of Natural Language Processing (NLP), enabling machines to understand and process human language with remarkable accuracy and efficiency. These models are trained on large-scale text corpora using unsupervised or self-supervised learning techniques, allowing them to learn complex linguistic patterns, syntactic structures, and semantic relationships inherent in natural language. Unlike traditional approaches that rely heavily on manual feature engineering, pre-trained models automatically learn rich and meaningful representations of text, making them highly effective for a wide range of NLP tasks, including text classification, sentiment analysis, question answering, and mental health detection.

Among the most influential pre-trained models, BERT (Bidirectional Encoder Representations from Transformers) marked a major breakthrough in NLP by introducing bidirectional context understanding. Unlike earlier models that processed text in a unidirectional manner (either left-to-right or right-to-left), BERT considers both directions simultaneously. This bidirectional approach allows the model to develop a deeper and more comprehensive understanding of sentence structure and meaning. BERT is based on the transformer architecture, which utilizes self-attention mechanisms to capture relationships between all words in a sentence, regardless of their position. This makes it highly effective for tasks that require contextual understanding, such as mental health text classification.

Another important aspect of pre-trained language models is their ability to leverage transfer learning. Since these models are pre-trained on massive datasets, they already possess a strong understanding of general language patterns. This knowledge can be transferred to specific tasks through fine-tuning, reducing the need for large labeled datasets. This is particularly beneficial in the mental health domain, where obtaining high-quality labeled data can be challenging due to privacy concerns and ethical considerations. Transfer learning enables the development of high-performance models even with limited data, making pre-trained models both practical and efficient.

Furthermore, pre-trained transformer models offer flexibility and scalability, allowing them to be adapted for various applications beyond classification, such as text generation, summarization, and conversational systems. Their ability to handle diverse linguistic inputs and generalize across domains makes them a powerful tool in modern NLP research.

Overall, pre-trained language models have revolutionized NLP by enabling more accurate, efficient, and

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scalable text processing. The transition from traditional feature-based methods to transformer-based architectures represents a major advancement in the field. Models such as BERT and RoBERTa have laid the foundation for developing intelligent systems capable of understanding complex human language. In this project, the adoption of RoBERTa-large, combined with domain-specific fine-tuning, provides a strong and effective framework for mental health text classification.

Among the most influential models, BERT (Bidirectional Encoder Representations from Transformers) introduced a major breakthrough by enabling bidirectional context understanding. This means that the model considers both the left and right context of a word simultaneously, leading to a deeper understanding of sentence meaning. This capability is particularly important in mental health analysis, where the meaning of a sentence often depends on subtle contextual cues. BERT demonstrated significant improvements over previous models in various NLP benchmarks, paving the way for more advanced architectures.

Additionally, pre-trained transformer models reduce the need for extensive labeled data, as they already possess a strong understanding of language from pre-training. This is particularly beneficial in domains like mental health, where labeled datasets may be limited or imbalanced. By leveraging transfer learning, these models can achieve high performance even with relatively smaller datasets, making them a practical and efficient solution for real-world applications.

2.4 HANDLING CLASS IMBALANCE

Class imbalance is one of the most critical challenges encountered in mental health text classification tasks, particularly when working with real-world datasets. In such datasets, certain categories such as *Normal* and *Depression* typically contain a large number of samples, while other categories like *Personality Disorder*, *Bipolar*, or *Suicidal* may have significantly fewer instances. This uneven distribution creates a bias during model training, where the learning algorithm tends to prioritize majority classes and neglect minority classes. As a result, the model may achieve high overall accuracy but perform poorly in identifying underrepresented categories, which are often the most important in mental health applications.

The impact of class imbalance is particularly severe in sensitive domains such as mental health detection, where misclassification of minority classes can have serious consequences. For example, failing to correctly identify suicidal or high-risk cases may prevent timely intervention. Therefore, it is essential to design models that are not only accurate but also fair and balanced across all classes. This requires the adoption of specialized techniques that ensure equal attention is given to both majority and minority classes during training.

In addition to focal loss, class weighting is another widely used technique to handle imbalance. In this approach, higher weights are assigned to minority classes and lower weights to majority classes during the computation of the loss function. This ensures that errors made on underrepresented classes have a greater impact on the model's learning process. As a result, the model becomes more sensitive to minority classes and achieves a more balanced performance.

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In recent research, additional strategies such as data augmentation and ensemble learning have also been explored to address class imbalance. Data augmentation techniques, including back-translation and paraphrasing, can be used to generate new samples for minority classes, thereby increasing dataset diversity. Ensemble methods combine predictions from multiple models to improve overall performance and reduce bias toward majority classes. Although these methods add computational complexity, they can significantly enhance classification accuracy in challenging scenarios.

Despite the availability of various techniques, handling class imbalance remains a complex problem that requires careful consideration of the dataset and task requirements. The choice of method depends on factors such as dataset size, degree of imbalance, and model architecture. In transformer-based models, such as RoBERTa, loss-based approaches like focal loss are often preferred due to their compatibility with large-scale training and ability to directly influence the learning process.

Furthermore, label smoothing is incorporated to prevent the model from becoming overly confident in its predictions. It distributes a small portion of probability mass across all classes, improving generalization and reducing overfitting. When combined with focal loss and class weighting, label smoothing enhances the overall robustness of the model. These techniques collectively ensure that the classification system performs effectively across all mental health categories, making it more reliable for real-world applications.

2.5 POOLING STRATEGIES

In Pooling strategies play a crucial role in extracting meaningful and informative representations from transformer-based models in Natural Language Processing (NLP). Transformer architectures, such as BERT and RoBERTa, generate contextual embeddings for each token in an input sequence. However, for downstream tasks like text classification, these token-level embeddings must be aggregated into a fixed-length vector that represents the entire input. The effectiveness of this aggregation process directly impacts the performance of the classification model, making pooling strategies an important area of research.

In many transformer-based models, the CLS (classification) token is commonly used as a representation of the entire input sequence. The CLS token is a special token added at the beginning of the input, and its corresponding embedding is designed to capture the overall meaning of the sentence. During training, the model learns to encode global contextual information into this token, making it suitable for classification tasks. While this approach is simple and computationally efficient, it has certain limitations. In complex tasks such as mental health text classification, relying solely on the CLS token may not capture all relevant information, especially when important features are distributed across different parts of the text.

Given the individual strengths and limitations of these methods, recent research has focused on combining multiple pooling strategies to achieve a more comprehensive representation. One such approach is triple pooling, which integrates CLS pooling, mean pooling, and max pooling into a single feature vector. By combining these three representations, the model benefits from global sentence-level context (CLS), overall semantic information (mean pooling), and important feature extraction (max pooling). This hybrid approach provides a richer and more balanced representation of the input text.

The use of triple pooling has been shown to significantly improve classification performance, particularly in tasks involving complex and nuanced data such as mental health text analysis. Mental health-related text often contains subtle linguistic variations, implicit meanings, and emotionally charged expressions that may not be fully captured by a single pooling method. By leveraging multiple pooling strategies, the model can better understand these nuances and differentiate between closely related mental health conditions.

Despite their advantages, combining multiple pooling strategies may increase computational complexity and model size. However, the performance gains achieved through improved feature representation often outweigh these costs, particularly in high-stakes applications such as mental health detection. Therefore, the adoption of advanced pooling techniques is considered a valuable enhancement in modern NLP systems.

In summary, pooling strategies are a fundamental component of transformer-based models, directly influencing their ability to extract meaningful representations from textual data. While traditional approaches rely on the CLS token, alternative methods such as mean and max pooling provide complementary strengths. The integration of these methods through triple pooling offers a comprehensive solution that captures global context, semantic meaning, and key features simultaneously. This approach enhances the accuracy, robustness, and reliability of classification systems, making it particularly suitable for complex tasks like mental health text analysis.

play a crucial role in extracting meaningful representations from transformer-based models. In most NLP tasks, the CLS token is commonly used as a representation of the entire input sequence. While this approach provides a global summary of the text, it may not always capture all the important information, especially in complex tasks such as mental health classification where subtle linguistic variations are significant.

Additionally, using multiple pooling strategies reduces the risk of information loss and improves the robustness of the model. It ensures that different aspects of the text are effectively captured, making the system more capable of handling diverse and unstructured data. This approach is particularly beneficial in mental health analysis, where different expressions may convey similar emotions. Therefore, triple pooling plays an important role in improving the performance and reliability of transformer-based classification systems.

2.6 DATA AUGMENTATION TECHNIQUES

Data augmentation plays a vital role in improving the performance and robustness of Natural Language Processing (NLP) models, particularly when working with limited, imbalanced, or noisy datasets. In the domain of mental health text classification, collecting large-scale, high-quality labeled data is inherently challenging due to privacy concerns, ethical constraints, and the sensitive nature of the content. As a result, available datasets are often small and unevenly distributed across different mental health categories. This limitation can negatively impact model performance, leading to overfitting and poor generalization. To address these issues, data augmentation techniques are employed to artificially increase the size and diversity of the training dataset, enabling the model to learn more comprehensive and generalized patterns.

One of the most widely used and effective data augmentation techniques in NLP is back-translation. In this approach, the original text is first translated into an intermediate language and then translated back into the original language. This process generates paraphrased versions of the input text while preserving its core meaning. The resulting variations introduce differences in sentence structure, word choice, and phrasing, which help the model learn diverse linguistic representations. Back-translation is particularly beneficial in mental health classification, where subtle differences in language, tone, and expression can significantly influence the interpretation of psychological states. By exposing the model to multiple variations of the same content, back-translation improves its ability to handle diverse expressions and enhances overall classification accuracy.

Another important advantage of data augmentation is its ability to address class imbalance. In many mental health datasets, minority classes such as *Bipolar*, *Suicidal*, or *Personality Disorder* have significantly fewer samples compared to majority classes. Augmentation techniques can be selectively applied to these underrepresented categories to generate additional training examples. This helps balance the dataset and ensures that the model is exposed to a wider range of samples from all classes. As a result, the model becomes more capable of accurately identifying minority class instances, improving both fairness and reliability.

Data augmentation also plays a crucial role in reducing overfitting. When a model is trained on a limited dataset, it may memorize specific patterns rather than learning generalizable features. By increasing dataset diversity, augmentation forces the model to learn broader patterns and relationships within the data. This leads to improved generalization performance when the model is applied to unseen or real-world inputs. In mental health applications, where variability in language is high, generalization is particularly important for ensuring consistent performance across different user inputs.

Recent advancements have also explored the use of contextual augmentation techniques, where pre-trained language models are used to generate new training samples. These methods leverage the contextual understanding of models like BERT or RoBERTa to create high-quality paraphrases that are both diverse and semantically consistent. Such approaches offer a more controlled and context-aware form of augmentation, further improving model performance.

Despite its advantages, data augmentation must be applied carefully to avoid introducing noise or altering the original meaning of the text. In mental health analysis, preserving the emotional and semantic integrity of the data is critical. Poorly generated augmented samples may mislead the model and negatively affect performance. Therefore, selecting appropriate augmentation techniques and validating the generated data are essential steps in the process.

In summary, data augmentation is a powerful strategy for enhancing the performance, robustness, and generalization capability of NLP models in mental health text classification. Techniques such as back-translation, synonym replacement, and paraphrasing help increase dataset diversity and address class imbalance. Additionally, augmentation improves the model's ability to handle noisy and unstructured text, making it more suitable for real-world applications. By incorporating effective augmentation strategies, the proposed system achieves improved accuracy and reliability, contributing to the development of a scalable and practical mental health detection framework.

In addition to back-translation, augmentation helps address class imbalance by generating more samples for underrepresented classes. This ensures that the model is exposed to a wider variety of examples from minority categories, improving its ability to correctly classify them. By increasing the diversity of training data, augmentation reduces the risk of overfitting and enhances the model's generalization capability when applied to real-world data.

Furthermore, data augmentation techniques contribute to improving the robustness of the model against noisy and unstructured text. Online textual data often contains slang, abbreviations, spelling errors, and informal language. Augmented data exposes the model to such variations, enabling it to perform better in real-world scenarios. Therefore, incorporating data augmentation strategies is essential for building a reliable and scalable mental health text classification system.

2.7 OPTIMIZATION TECHNIQUES

Optimization techniques play a crucial role in effectively training deep learning models, particularly in transformer-based architectures used for mental health text classification. These models, such as RoBERTa-large, consist of millions of parameters and require careful tuning to achieve optimal performance. Proper optimization ensures that the model converges efficiently during training while maintaining numerical stability and avoiding common issues such as vanishing gradients, exploding gradients, and overfitting. The choice of optimization strategies directly impacts the model's accuracy, generalization capability, and training efficiency.

One of the most widely used optimization algorithms in modern deep learning is AdamW (Adaptive Moment Estimation with Weight Decay). AdamW extends the traditional Adam optimizer by decoupling weight decay from the gradient-based update, allowing more effective regularization. This helps prevent overfitting by penalizing large weights while still benefiting from adaptive learning rates for each parameter. In transformer-based models, AdamW has become the standard optimizer due to its ability to handle large-scale parameter updates efficiently and maintain stable convergence during training.

In addition to selecting an appropriate optimizer, learning rate scheduling plays a critical role in model optimization. The learning rate determines how much the model parameters are updated during each training step. If the learning rate is too high, the model may become unstable and fail to converge; if it is too low, training may become slow and inefficient. To address this, advanced scheduling techniques such as cosine annealing with warm-up are employed. During the warm-up phase, the learning rate gradually increases from a small value to a predefined maximum, allowing the model to stabilize in the early stages of training. After this phase, the learning rate is gradually decreased following a cosine decay pattern, enabling fine-tuning of model parameters and improved convergence.

Another important optimization technique is gradient clipping, which is used to prevent the problem of exploding gradients during backpropagation. In deep neural networks, especially those with many layers, gradients can sometimes grow excessively large, leading to unstable updates and divergence during training. Gradient clipping addresses this issue by setting a threshold and scaling down gradients that exceed this limit. This ensures smoother and more stable training, particularly when dealing with complex models and large datasets.

Furthermore, layer-wise learning rate decay is an important optimization strategy specifically designed for transformer-based models. In this approach, different layers of the model are assigned different learning rates based on their position. Lower layers, which capture general linguistic knowledge from pre-training, are updated with smaller learning rates to preserve their learned representations. In contrast, higher layers, which are more task-specific, are updated with larger learning rates to adapt to the target domain. This selective updating helps in balancing knowledge retention and domain adaptation, resulting in improved performance and reduced risk of overfitting.

Regularization techniques also play a significant role in optimization. Methods such as dropout are used to randomly deactivate a subset of neurons during training, preventing the model from relying too heavily on specific features. This encourages the model to learn more generalized patterns and improves its ability to perform well on unseen data. In transformer models, dropout is applied at various stages, including attention layers and feed-forward networks, contributing to improved robustness.

Batch size selection is another factor that influences optimization. Larger batch sizes can lead to more stable gradient estimates and faster training, but they require more computational resources. Smaller batch sizes, on the other hand, introduce more noise in gradient updates, which can sometimes help in escaping local minima. Therefore, selecting an appropriate batch size is essential for achieving a balance between training efficiency and model performance.

Early stopping is also commonly used as a practical optimization strategy. During training, the model's performance on a validation set is monitored, and training is stopped when performance no longer improves. This prevents overfitting and reduces unnecessary computation, ensuring that the model retains its best-performing state.

Overall, optimization techniques are fundamental to the successful training of deep learning models for mental health text classification. The combination of advanced optimizers such as AdamW, learning rate scheduling strategies, gradient clipping, mixed precision training, and layer-wise learning rate decay provides a robust framework for efficient and stable model training. These techniques ensure that the model not only achieves high accuracy but also maintains reliability and generalization capability when applied to real-world data. By carefully integrating these optimization strategies, the proposed system is able to effectively leverage the power of transformer-based models and deliver high-performance results in mental health analysis.

One of the most widely used optimizers in deep learning is AdamW, which combines the benefits of adaptive learning rates with weight decay regularization. This helps in controlling overfitting and improving generalization. In addition, learning rate scheduling techniques such as cosine annealing with warm-up are applied to gradually adjust the learning rate during training. This allows the model to learn efficiently in the initial stages and fine-tune its parameters in later stages.

Furthermore, layer-wise learning rate decay is applied in transformer models to assign different learning rates to different layers. Higher layers are trained with larger learning rates, while lower layers are updated more conservatively. This approach helps in preserving pre-trained knowledge while adapting the model to the specific task, resulting in improved performance and stability.

2.8 PROBABILITY CALIBRATION

Probability calibration is a critical aspect of machine learning models, particularly in sensitive application domains such as mental health analysis, where the reliability and interpretability of predictions are of utmost importance. In classification tasks, models typically output probability scores that represent the confidence of predictions for each class. Ideally, these probabilities should reflect the true likelihood of correctness. However, modern deep learning models, especially large transformer-based architectures, often produce overconfident predictions. This means that the predicted probabilities are disproportionately high even when the model is uncertain, leading to a mismatch between confidence scores and actual performance.

Overconfidence in predictions can significantly reduce trust in the system and may lead to incorrect interpretations, particularly in real-world applications. In the context of mental health classification, this issue becomes even more critical, as decisions or insights derived from model outputs may influence further actions such as screening, monitoring, or intervention. Therefore, ensuring that the predicted probabilities are well-calibrated is essential for building reliable and trustworthy systems.

To address this issue, probability calibration techniques are employed to align predicted confidence scores with true outcome probabilities. One of the most widely used and effective methods for calibration is temperature scaling, which is applied as a post-training procedure. In this technique, the logits (raw output values) produced by the model are divided by a scalar parameter known as the temperature before applying the softmax function. The temperature parameter is learned using a validation dataset, and it adjusts the sharpness of the probability distribution.

Calibration techniques significantly improve the interpretability of model outputs. Well-calibrated probabilities provide users with a clearer understanding of how confident the model is in its predictions. For example, if a model predicts a class with 80% confidence, a well-calibrated system ensures that approximately 80% of such predictions are correct. This level of reliability is essential in mental health applications, where users and practitioners rely on the system to provide meaningful and trustworthy insights.

Furthermore, probability calibration helps in reducing the risks associated with incorrect predictions. In many cases, uncalibrated models may assign high confidence to wrong predictions, which can lead to misleading conclusions. By adjusting the confidence scores, calibration techniques ensure that the model expresses uncertainty more accurately. This allows users to make informed decisions and exercise caution when dealing with low-confidence predictions.

In summary, probability calibration is an essential component of modern machine learning systems, particularly in high-stakes domains such as mental health analysis. Techniques such as temperature scaling help align predicted probabilities with actual outcomes, improving both reliability and interpretability. By reducing overconfidence and providing balanced confidence scores, calibration enhances the trustworthiness of the system and supports more informed decision-making. Incorporating effective calibration strategies ensures that the classification model is not only accurate but also dependable and suitable for real-world deployment.

Additionally, probability calibration helps in reducing the risk of overconfidence in incorrect predictions. By providing balanced confidence scores, the system becomes more trustworthy and suitable for deployment in sensitive domains. Therefore, incorporating calibration techniques is essential for building robust and dependable classification systems.

2.9 EXISTING SYSTEM LIMITATIONS

Despite the significant advancements in Natural Language Processing (NLP) and deep learning techniques, existing mental health text classification systems continue to face several important limitations. These challenges affect the accuracy, reliability, and real-world applicability of such systems, particularly in sensitive domains like mental health analysis. Understanding these limitations is essential for identifying research gaps and motivating the development of more robust and effective approaches.

Although modern transformer-based models have improved contextual understanding, challenges still remain in effectively modelling long and complex text sequences. In many cases, important contextual cues may be distributed across different parts of the text, making it difficult for models to fully capture all relevant information. This limitation can affect the system’s ability to detect subtle indicators of mental health conditions, particularly when dealing with ambiguous or indirect expressions.

Another important limitation is the overlapping nature of mental health conditions. Different psychological disorders often share similar symptoms and linguistic patterns, making it difficult for models to distinguish between them. For example, expressions of sadness, hopelessness, or fatigue may appear in both depression and anxiety-related text. This overlap creates ambiguity in classification and increases the complexity of the task. Many existing models lack the capability to effectively differentiate between closely related classes, resulting in misclassification.

Ethical and practical challenges also play a significant role in limiting the adoption of these systems. Concerns related to data privacy, consent, and potential misuse of sensitive information must be carefully addressed. Additionally, most existing systems are developed for research purposes and may not be fully validated for clinical use. This limits their applicability in real-world healthcare settings.

These limitations highlight the need for improved approaches that can effectively handle complex and noisy data, address class imbalance, reduce overfitting, and provide reliable and interpretable predictions. Advanced techniques such as transformer-based models, improved pooling strategies, data augmentation, loss function optimization, and probability calibration offer promising solutions to these challenges. Addressing these issues is essential for developing robust and scalable mental health classification systems that can be safely and effectively used in practical applications.

the advancements in NLP and deep learning, existing mental health classification systems still face several limitations. One of the major challenges is the inability of traditional models to capture deep contextual relationships within text. Earlier approaches relied heavily on manual feature extraction, which limited their effectiveness in handling complex language patterns.

Another significant limitation is the issue of class imbalance, where some mental health categories are underrepresented in the dataset. Many existing systems fail to address this problem effectively, resulting in biased predictions. Additionally, large deep learning models often suffer from overfitting, especially when trained on moderately sized datasets, reducing their performance on unseen data.

2.10 SUMMARY

The The literature review presented in this chapter highlights the significant advancements in the field of mental health text classification, demonstrating a clear evolution from traditional machine learning approaches to modern deep learning and transformer-based models. Early methods, which relied on manual feature engineering techniques such as Bag-of-Words and TF-IDF, provided basic insights into textual data but were limited in their ability to capture deep contextual relationships and semantic meaning. These limitations made it difficult to effectively analyse complex and nuanced language patterns commonly found in mental health-related text.

With the introduction of neural embeddings and deep learning architectures, the field experienced a substantial improvement in performance. Models such as Recurrent Neural Networks (RNNs) and Long

Short-Term Memory (LSTM) networks enabled better handling of sequential data and contextual dependencies. However, the most significant breakthrough came with the development of transformer-based models, which utilize self-attention mechanisms to capture relationships between words more effectively. Among these, models like RoBERTa have demonstrated superior performance due to their optimized training strategies and ability to process large-scale data. These models have set new benchmarks in NLP tasks, including mental health classification, by providing a deeper understanding of language context and semantics.

To address these challenges, various advanced techniques have been explored in the literature. Methods such as focal loss, class weighting, and label smoothing have been shown to effectively handle class imbalance and improve model generalization. Data augmentation techniques, particularly back-translation, play a crucial role in increasing dataset diversity and enhancing model robustness. Furthermore, advanced feature representation strategies, such as triple pooling, enable the model to capture multiple aspects of textual information, leading to improved classification accuracy.

Overall, the literature provides a strong theoretical and practical foundation for the proposed system developed in this project. It highlights the importance of adopting advanced NLP techniques and transformer-based architectures while addressing critical challenges such as class imbalance, overfitting, and data variability. By building upon these insights, the proposed system aims to overcome the limitations of existing approaches and deliver improved performance in real-world mental health analysis applications. This comprehensive understanding of prior research ensures that the proposed methodology is both well-informed and aligned with current advancements in the field.

The review also emphasizes the importance of addressing key challenges such as class imbalance, overfitting, and noisy data. Techniques like focal loss, label smoothing, and data augmentation have been shown to improve model performance and robustness. Additionally, advanced strategies such as triple pooling and probability calibration enhance feature representation and prediction reliability.

From the analysis of existing research, it is clear that combining multiple techniques leads to more accurate and reliable systems. The integration of optimization strategies, calibration methods, and data augmentation plays a crucial role in improving classification performance.

Overall, the reviewed literature provides a strong basis for designing a scalable and efficient mental health classification model. By leveraging advanced NLP techniques and transformer-based architectures, the proposed system aims to overcome existing limitations and deliver improved performance in real-world applications.

2.11 TRANSFER LEARNING IN NLP

Transfer learning has emerged as one of the most transformative paradigms in Natural Language Processing, fundamentally changing how models are developed and deployed for downstream tasks. The central idea behind transfer learning is to leverage knowledge acquired from training on large-scale general-purpose corpora and apply it to specialized tasks with limited labeled data. This approach has proven particularly valuable in domains such as mental health text analysis, where the collection of large annotated datasets is often infeasible due to ethical constraints and data scarcity.

The pretraining and fine-tuning paradigm, popularized by BERT and its successors, involves two stages. In the pretraining stage, the model is trained on massive unlabeled text corpora using self-supervised objectives such as masked language modeling (MLM) and next sentence prediction (NSP). This allows the model to develop deep representations of language that capture syntactic structure, semantic relationships, and contextual meaning. In the fine-tuning stage, the pretrained model is adapted to the target task by training on a smaller labeled dataset, enabling it to leverage the general linguistic knowledge acquired during pretraining.

In the context of mental health text classification, transfer learning enables the development of high-performing models even when the available labeled mental health dataset is relatively small compared to the pretraining corpus. The rich linguistic representations learned during pretraining provide the model with a strong foundation for understanding complex emotional and psychological language, reducing the amount of domain-specific data required to achieve good performance. This is a significant advantage given the practical challenges of collecting large-scale clinical mental health datasets.

2.12 ATTENTION MECHANISMS AND SELF-ATTENTION IN TRANSFORMERS

The self-attention mechanism forms the core computational foundation of transformer-based architectures and is one of the most significant innovations in modern Natural Language Processing (NLP). It enables models to process and understand textual data in a highly efficient and context-aware manner. Unlike traditional sequential models such as Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks, which process tokens one at a time, self-attention allows all tokens in a sequence to be processed simultaneously. This parallel processing capability not only improves computational efficiency but also enables the model to capture long-range dependencies within text more effectively.

The self-attention operation is based on three key components: query (Q), key (K), and value (V) vectors. For each token in the input sequence, these vectors are generated through learned linear transformations. The attention score between two tokens is computed using the scaled dot-product of their query and key vectors. These scores are then normalized using the softmax function to produce attention weights, which represent the importance of each token relative to others. The final representation of each token is obtained by computing a weighted sum of the value vectors, where the weights are determined by the attention scores.

Mathematically, the self-attention operation can be expressed as:

$$\text{Attention}(Q, K, V) = \text{SoftMax}((QK^T) / \sqrt{d_k})\, V$$

where d_k represents the dimensionality of the key vectors. The scaling factor $\sqrt{d_k}$ is used to prevent

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excessively large values during the dot-product computation, ensuring numerical stability during training.

To further enhance the model's ability to capture diverse relationships within text, transformer architectures employ multi-head attention. In this approach, multiple attention mechanisms (referred to as "heads") are applied in parallel. Each head learns to focus on different aspects of the input, such as syntactic structure, semantic meaning, or positional relationships. The outputs from all heads are then concatenated and linearly transformed to produce the final representation. This allows the model to capture multiple perspectives of the input simultaneously, resulting in richer and more informative representations.

In the context of mental health text analysis, the self-attention mechanism is particularly valuable due to the complex and nuanced nature of psychological language. Expressions related to mental health often involve implicit meanings, emotional subtleties, and long-range dependencies. For example, a sentence may begin with a neutral tone but later include words that indicate distress or negativity. Traditional models may struggle to capture such dependencies, whereas self-attention can effectively identify and relate these patterns.

In summary, the self-attention mechanism represents a fundamental advancement in NLP, enabling models to capture long-range dependencies, contextual relationships, and nuanced linguistic patterns. Its integration into transformer architectures has significantly improved performance across a wide range of tasks, including mental health text classification. By leveraging self-attention, the proposed system is able to achieve a deeper understanding of textual data, leading to more accurate and reliable predictions. self-attention mechanism is the core computational building block of transformer architectures and is fundamental to their ability to capture long-range dependencies in text

2.13 RELATED WORK ON SUICIDE AND SELF-HARM DETECTION

The automated detection of suicidal ideation and self-harm risk from textual data has emerged as a critical research area within the broader domain of mental health analysis using Natural Language Processing (NLP). With the increasing prevalence of online platforms, individuals often express distress, emotional struggles, and suicidal thoughts through text, making it possible to analyse such data for early detection and intervention. This has led to the development of various computational approaches aimed at identifying high-risk individuals based on their language patterns.

Early research in this area primarily relied on keyword-based methods and lexicon matching techniques. These approaches used predefined dictionaries of words and phrases associated with suicide and self-harm to identify relevant content. While these methods were simple and computationally efficient, they lacked the ability to understand context. As a result, they often produced high rates of false positives and false negatives. For example, a sentence mentioning the word "die" in a non-suicidal context could be incorrectly flagged, while subtle expressions of distress without explicit keywords might be missed.

However, they still struggled to generalize across diverse datasets and were limited in their ability to capture deep semantic relationships.

The introduction of benchmark datasets and shared tasks significantly accelerated research in this area. Notably, the CLPsych shared tasks conducted between 2015 and 2019 provided standardized datasets and evaluation frameworks for mental health detection systems. These initiatives encouraged the development and comparison of different approaches, leading to steady improvements in model performance.

In recent years, the adoption of deep learning and transformer-based models has revolutionized suicide and self-harm detection. Models such as BERT and RoBERTa, when fine-tuned on large datasets of social media posts, have demonstrated substantial improvements in performance. These models are capable of capturing contextual meaning, emotional tone, and subtle linguistic patterns that are often indicative of suicidal ideation. Unlike earlier methods, transformer-based models can understand the broader context of a sentence, reducing the likelihood of misclassification.

Studies using data from platforms such as Reddit, Twitter, and online support forums have reported macro F1-scores exceeding 90% for binary classification tasks involving suicidal ideation detection. These results highlight the effectiveness of pre-trained language models in analyzing emotionally complex and sensitive text. Additionally, these models have shown the ability to generalize across different domains, making them suitable for real-world applications.

Subsequent research applied machine learning classifiers such as Support Vector Machines and Logistic Regression with engineered features including sentiment scores, linguistic markers, and psycholinguistic word counts derived from tools such as LIWC (Linguistic Inquiry and Word Count). These approaches demonstrated improved performance over keyword matching but remained limited in their ability to generalize to diverse linguistic expressions of suicidal distress. The CLPsych shared tasks organized between 2015 and 2019 provided standardized benchmarks for evaluating automated mental health detection systems, fostering significant research activity in this space.

More recent work has demonstrated that transformer-based models fine-tuned on large collections of social media posts related to suicide and self-harm achieve substantially higher performance than earlier approaches. Models such as BERT and RoBERTa, when fine-tuned on datasets from platforms like Reddit, have achieved macro F1-scores exceeding 90% on binary suicidal ideation detection tasks.

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CHAPTER 3

PROPOSED METHOD

This chapter presents the overall methodology adopted for developing the proposed mental health text classification system. The primary objective of the methodology is to design a robust and efficient pipeline that can process user-generated textual data and accurately classify it into predefined mental health categories. The system is built using advanced Natural Language Processing (NLP) techniques combined with deep learning models to capture complex linguistic patterns and contextual relationships present in the text.

The methodology consists of several stages, including data preprocessing, tokenization, feature extraction, model training, and evaluation. Each stage plays a critical role in ensuring the overall performance and reliability of the system. Special attention is given to handling challenges such as class imbalance, noisy data, and overlapping language patterns, which are common in mental health datasets. By incorporating appropriate techniques, the system aims to achieve high accuracy and balanced performance across all categories.

Furthermore, the proposed system leverages the RoBERTa-large transformer model, which is known for its strong contextual understanding capabilities. The model is fine-tuned on domain- specific data to adapt it for mental health classification. Additional techniques such as triple pooling, focal loss, and temperature scaling are integrated to enhance feature representation and prediction reliability.

Another important design consideration is the prioritization of model interpretability and deployment readiness alongside raw classification performance. The integration of temperature-scaled probability calibration ensures that the model's confidence scores are meaningful and actionable, rather than arbitrarily high values that give a false impression of certainty. The development of the Streamlit-based web application further reflects the commitment to making the system practically usable rather than remaining an academic prototype. Together, these design choices reflect a holistic approach to system development that considers not only technical performance metrics but also the practical requirements and ethical responsibilities associated with deploying AI systems in sensitive healthcare-adjacent domains.

METHODOLOGY

3.1 PROBLEM FORMULATION

Given an input text sequence $x = (x_1, x_2, . . . , x_n)$, the objective of this study is to predict the corresponding mental health category $y \in \{1,2, . . . ,7\}$, representing the classes: Normal, Depression, Suicidal, Anxiety, Bipolar, Stress, and Personality Disorder. This task is formulated as a supervised multi-class text classification problem, where each input instance is mapped to one of the predefined categories. The formulation allows the model to learn patterns from labeled data and generalize its predictions to unseen textual inputs.

The proposed model estimates a conditional probability distribution $\hat{p}(y \mid x) \in \mathbb{R}^7$, which represents the likelihood of the input text belonging to each mental health class. The final prediction is obtained by applying a temperature-scaled softmax function to the model logits, ensuring calibrated probability outputs. This probabilistic framework enables the system to not only classify text but also provide confidence levels associated with each prediction.

To address these challenges, the problem formulation incorporates contextual representation learning using transformer models, which capture semantic relationships within the text. By leveraging deep contextual embeddings and calibrated probability estimation, the system is capable of performing robust and reliable mental health classification suitable for real-world applications

One of the key challenges in this formulation is the inherent ambiguity in natural language, especially in mental health contexts. A single sentence may contain overlapping indicators of multiple conditions, making classification complex. Additionally, variations in writing style, emotional tone, and linguistic expression further increase the difficulty of accurately mapping text to specific categories.

To address these challenges, the problem formulation incorporates contextual representation learning using transformer models, which capture semantic relationships within the text. By leveraging deep contextual embeddings and calibrated probability estimation, the system is capable of performing robust and reliable mental health classification.

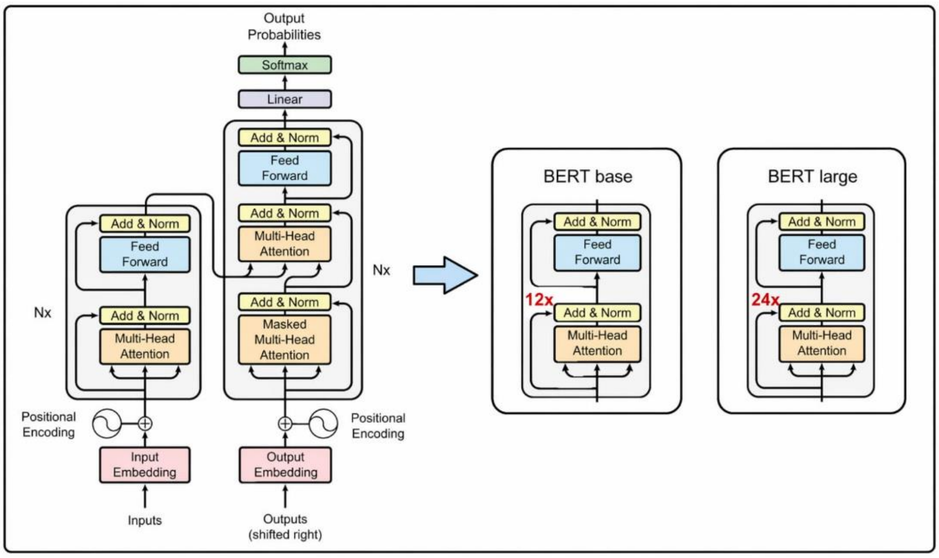

Figure-3.1.1: BERT Architecture

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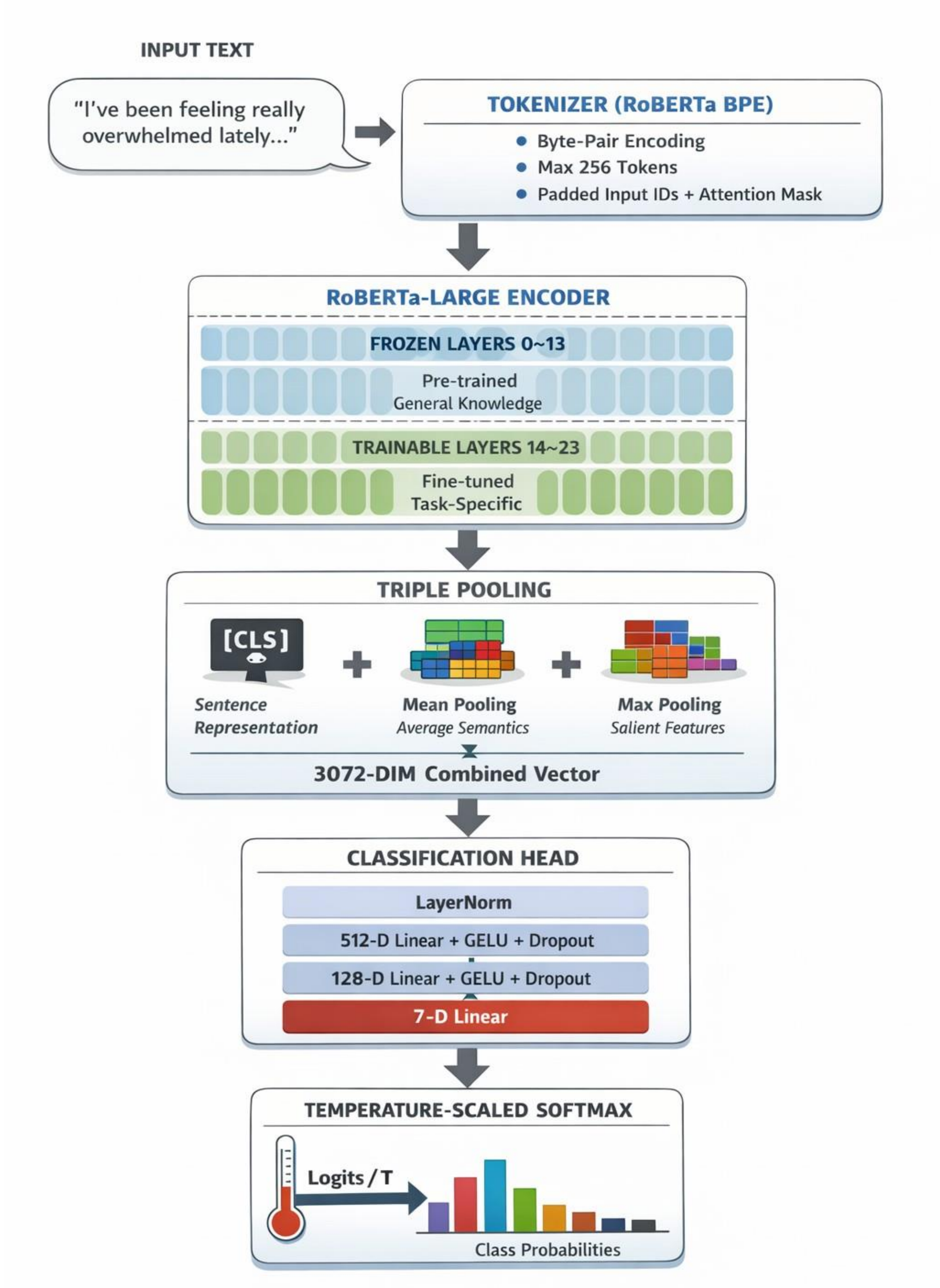

Figure 3.1.2: Methodology

This Steps of the Proposed System Architecture:

- **Input text**
- **Tokenization (RoBERTsa BPE)**
- **RoBERTa-Large Encoder**
- **Triple pooling**
- **Classification Head**
- **Temperature -Scaled Softmax**
- **Output Prediction**

3.2 ARCHITECTURE OVERVIEW:

The proposed system is based on a transformer-based architecture that processes textual input through multiple stages to generate accurate predictions. The architecture consists of key components such as tokenization, encoding, feature extraction, and classification. Each component is designed to perform a specific function and contributes to the overall performance of the system.

The input text is first processed using a tokenizer that converts it into a sequence of tokens. These tokens are then passed to the RoBERTa-large encoder, which generates contextual embeddings by capturing relationships between words. The encoder plays a crucial role in understanding the meaning of the text and extracting relevant features.

To enhance feature representation, a triple pooling mechanism is applied, which combines different pooling strategies to capture multiple aspects of the input text. The resulting feature vector is then passed through a classification head consisting of fully connected layers. This component transforms the extracted features into class probabilities.

The architecture is designed to be both efficient and scalable, allowing it to handle large datasets and complex inputs. By integrating multiple advanced techniques, the system achieves high performance and reliability in mental health classification tasks.

➢ Tokenization

The input text is first processed using RoBERTa's Byte-Pair Encoding (BPE) tokenizer, which efficiently handles sub word representations and out-of-vocabulary terms. A maximum sequence length of 256 tokens is enforced, and shorter sequences are padded while longer sequences are truncated. This ensures consistent input dimensions across all samples. Attention masks are generated to distinguish valid tokens from padding, enabling the model to focus only on meaningful textual content during processing**.**

➢ Encoder — RoBERTa-large

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3.2.1 TOKENIZATION AND INPUT REPRESENTATION

Tokenization is the foundational step that converts raw text into a format compatible with the RoBERTa-large model. The Byte-Pair Encoding (BPE) algorithm used by RoBERTa operates by iteratively merging the most frequently co-occurring pairs of characters or character sequences in the training corpus until a target vocabulary size is reached. The RoBERTa tokenizer employs a vocabulary of 50,265 unique tokens, which includes both complete words and frequent sub word units. This sub word representation allows the tokenizer to handle rare and out-of-vocabulary words effectively by decomposing them into known sub word components, ensuring that the model can process diverse and informal text such as social media posts without encountering unknown tokens.

Each input text is tokenized and formatted as a sequence beginning with a special CLS token and ending with a special SEP token. The CLS token plays a dual role in the model: it serves as a placeholder that aggregates global sequence-level information through the self-attention mechanism, and its final hidden state is conventionally used as the sentence-level representation for classification tasks. The SEP token serves as a delimiter marking the end of the input sequence. For the triple pooling strategy used in this study, the hidden states of all tokens including CLS and SEP are used to compute the mean and max pooling representations, providing a more comprehensive view of the encoded sequence than the CLS representation alone.

The maximum sequence length is set to 512 tokens, which is the maximum supported by RoBERTa-large. Texts that exceed this limit are truncated using a head-and-tail truncation strategy, which retains the first 128 tokens and the last 382 tokens of the sequence. This approach preserves both the beginning and end of longer texts, which often contain the most contextually relevant information for mental health classification. Shorter texts are padded with a special padding token to reach the maximum length, and attention masks are constructed to distinguish meaningful tokens (mask value 1) from padding tokens (mask value 0), ensuring that the encoder does not attend to padding positions during computation.

3.2.2 CLASSIFICATION HEAD DESIGN

The classification head is the task-specific component of the model that takes the pooled feature representation as input and produces a probability distribution over the seven mental health categories. It consists of a linear projection layer followed by a dropout layer and a final output linear layer. The input dimension of the classification head is 3072, which corresponds to the concatenation of three 1024-dimensional pooled representations from CLS, mean, and max pooling. The dropout probability is set to 0.1 to provide regularization and prevent overfitting within the classification head during training.

The encoder is based on the RoBERTa-large architecture, which consists of 24 transformer layers with a hidden size of 1024 and approximately 355 million parameters. The model is pre- trained on large-scale corpora and is capable of capturing deep contextual representations of language.

RoBERTa (Robustly Optimized BERT Pretraining Approach) was proposed by Liu et al. (2019) as an improved version of BERT with several key modifications to the pretraining procedure. First, RoBERTa removes the next sentence prediction (NSP) objective used in BERT, which was found to be detrimental to downstream task performance. Second, RoBERTa trains on significantly larger datasets, including the Books Corpus, English Wikipedia, CC-News, OpenWebText, and Stories datasets, totaling over 160 GB of text. Third, RoBERTa trains with larger batch sizes and longer sequences, and uses dynamic masking rather than static masking during pretraining, meaning the masking pattern is generated fresh each time a sequence is fed into the model. These improvements result in substantially better downstream performance compared to the original BERT model.

The RoBERTa-large variant used in this study consists of 24 transformer encoder layers, each with 16 self-attention heads and a feed-forward network of hidden dimension 4096. The input token embeddings have a dimension of 1024, and the model uses a Byte-Pair Encoding (BPE) vocabulary of 50,265 tokens. The total parameter count is approximately 355 million, making it one of the largest and most powerful transformer encoder models available for NLP tasks

The partial fine-tuning strategy adopted in this study involves freezing the lower 14 transformer layers (layers 0 to 13) of the RoBERTa-large encoder, which are responsible for learning general linguistic features such as syntax, morphology, and basic semantics. Only the upper 10 layers (layers 14 to 23) are fine-tuned during training, along with the classification head. This approach has two key advantages. First, it reduces the number of trainable parameters from approximately 356 million to about 127.6 million (35.8% of the total), significantly reducing the risk of overfitting on the relatively small mental health dataset. Second, it reduces the computational cost of training and allows the model to converge faster while still achieving strong performance.

Layer-wise learning rate decay is applied to the trainable layers to prevent catastrophic forgetting of the pretrained representations. Specifically, the learning rate decreases exponentially as we move from the upper layers toward the lower layersThis differential update strategy helps the model adapt effectively to the mental health domain while preserving the general linguistic knowledge encoded in the lower layers during pretraining.

To optimize performance and reduce computational overhead, the native pooling layer is removed. A partial fine-tuning strategy is adopted to balance knowledge retention and domain adaptation:

- The lower layers (0–13) are frozen to preserve general linguistic knowledge.
- The upper layers (14–23) are fine-tuned to capture domain-specific patterns related to mental health.
- The embedding layer remains frozen to maintain stable token-level representations.
- This approach reduces the number of trainable parameters to approximately **127.6 million**, thereby mitigating overfitting and improving generalization on limited datasets.

➢ Triple Pooling

This Instead of relying solely on the CLS token representation, a triple pooling mechanism is employed to capture diverse aspects of the input text. Let $H \in \mathbb{R}^{n\times 1024}$denote the final hidden states of the encoder.

$$v_{CLS} = [0] \in \mathbb{R}^{1024}$$

$$v_{mean} = \frac{\sum_{i=1}^{n} m_i . H[i]}{\sum_{i=1}^{n} m_i}$$

$$v_{max} = \max{}_{i:m_i=1}[i]$$

$$v_{pool} = [v_{CLS}; v_{mean}; v_{max]} \in \mathbb{R}^{3072}$$

Here, $m_i$represents the attention mask for token i. The CLS vector captures global sentence-level semantics, mean pooling captures overall contextual information, and max pooling extracts the most salient features. Numerical stability is ensured through epsilon clamping and masking strategies during pooling operations.

➢ Classification Head

The pooled feature vector is passed through a three-layer fully connected neural network:

$$z_1 = Dropout_{0.30}(GELU\ (W_1 \cdot LayerNorm(v_{pool}) + b_1))$$

$$z_2 = Dropout_{0.15}(GELU\ (W_2 \cdot z_1 + b_2))$$

$$\hat{y} = W_3 \cdot z_2 + b_3$$

Where:

- $W_1 \in \mathbb{R}^{512\times 3072}$
- $W_2 \in \mathbb{R}^{128\times 512}$
- $W_3 \in \mathbb{R}^{7\times 128}$

The use of GELU activation enhances non-linearity, while dropout regularization (0.30 → 0.15) reduces overfitting. Layer normalization stabilizes the concatenated feature representation before projection.

3.3 LOSS FUNCTION – FOCAL LOSS WITH LABEL SMOOTHING

To effectively address class imbalance, focal loss is employed with label smoothing:

$$\mathcal{L}_{focal} = -\frac{1}{N}\sum_{i=1}^{N} \alpha_{y_i} . \left(1 - p_{t,i}\right)^{\gamma} . CE_{smooth}(\hat{y}_i, y_i)$$

Where:

- $\boldsymbol{p_{t,i}}$ is the predicted probability for the true class
- $\boldsymbol{\gamma = 2.0}$ controls the focusing effect
- $\boldsymbol{\alpha_{y_i}}$ is the class weight based on inverse square root frequency The smoothed target distribution is defined as:

$$q_c = \begin{cases} 1-\epsilon & if\ c = y \\ \dfrac{\epsilon}{C-1} & Otherwise \end{cases}$$

with $\boldsymbol{\epsilon = 0.05}$.

This combination reduces overconfidence, improves generalization, and ensures that minority classes receive sufficient attention during training.

Once the input is collected, it is passed into the preprocessing module which performs operations such as cleaning, tokenization, normalization, and padding. This ensures that the text is converted into a structured format suitable for model processing

3.4 OPTIMIZATION STRATEGY

1 Layer-wise Learning Rate Decay:

Where:

3.4.1 $\eta_{bert} = 2 \times 10^{-5}$

3.4.2 $\gamma = 0.9$

This strategy assigns progressively smaller learning rates to lower layers, preserving pre-trained knowledge while allowing higher layers to adapt effectively. The classification head is trained with a higher learning rate $\eta_{head} = 1 \times 10^{-4}$.

Table 1: Dataset Class Distribution (After Filtering)

Table 1 presents the distribution of samples across the seven mental health categories after preprocessing and filtering of the dataset. It clearly illustrates that the dataset is not uniformly distributed, with certain classes dominating the overall sample count. In particular, the Normal and Depression categories contribute the largest portion of the dataset, followed by Suicidal instances, while classes such as Anxiety, Bipolar, Stress, and Personality Disorder have significantly fewer samples. This uneven distribution reflects the real- world scenario where some mental health conditions are more commonly reported than others.

The presence of such class imbalance introduces a critical challenge in the model training process. Machine learning models trained on imbalanced datasets tend to become biased toward majority classes, as they are exposed to them more frequently during training. As a result, the model may achieve high overall accuracy but fail to correctly identify minority class instances, which are often more critical in mental health applications. This imbalance can lead to poor recall and F1-scores for underrepresented classes, thereby reducing the effectiveness and reliability of the system.

Furthermore, the imbalance also affects the learning dynamics of deep neural networks, especially large transformer-based models. These models may prioritize dominant patterns present in majority classes while ignoring subtle features associated with minority categories. In the context of mental health classification, this is particularly problematic, as minority classes such as Personality Disorder or Bipolar conditions may carry significant clinical importance. Therefore, addressing class imbalance is essential not only for improving performance metrics but also for ensuring fair and meaningful predictions.

Class	Total Samples	Strategy	Train Split	Test Split
Anxiety	15,000	Downsampled	11,250	1,875
Bipolar	10,000	Oversampled	7,500	1,250
Depression	15,000	Downsampled	11,250	1,875
Normal	15,000	Downsampled	11,250	1,875
Personality Disorder	10,000	Oversampled	7,500	1,250
Stress	10,000	Oversampled	7,500	1,250
Suicidal	15,000	Downsampled	11,250	1,875

Figure-3.4.1: Dataset and Class Distribution

To mitigate these issues, the proposed system integrates several advanced techniques within the training pipeline. Focal loss is used to emphasize difficult and minority class samples, while class weighting ensures that underrepresented categories receive higher importance during optimization. Additionally, data augmentation techniques such as back-translation are employed to artificially increase the diversity and size of minority class samples. These combined strategies help in achieving balanced learning and improving generalization across all classes. Thus, the dataset distribution shown in Table 3.1 justifies the need for these enhancements and highlights their importance in the overall methodology.

class	**samples**	**percentage**
Normal	~16,150	30.78%
Depression	~15,400	29.35%
Suicidal	~10,650	20.30%
Anxiety	~3,840	7.32%
Bipolar	~2780	5.30%
Stress	~2590	4.97%
Personal Disorder	~1070	2.04%
Total	52,475	100%

Table 3.4.2: Dataset Class Distribution (After Filtering)

This distribution highlights the significant class imbalance, which motivates the use of specialized loss functions.

2. Training Procedure

The model is trained using the following configuration:

- Optimizer: AdamW with weight decay $\lambda = 0.02$
- Learning rate scheduler: Cosine annealing with 6% warmup
- Gradient accumulation: 4 steps (effective batch size = 64)
- Mixed precision training (FP16) for computational efficiency
- Gradient clipping with maximum norm = 1.0
- Early stopping based on validation macro F1-score
- Maximum training epochs: 20

These strategies collectively improve convergence speed, stability, and generalization performance.

3.5 POST-TRAINING TEMPERATURE SCALING

After the training phase, temperature scaling is applied as a post-processing step to improve the calibration of predicted probabilities. Deep learning models, especially large transformer- based architectures, often produce overconfident predictions, where the assigned probabilities do not accurately reflect the true likelihood of correctness. This issue can be critical in sensitive domains such as mental health analysis, where reliable confidence estimates are essential for meaningful interpretation. Therefore, probability calibration becomes an important step in enhancing the trustworthiness of the model.

Temperature scaling is a simple yet effective calibration technique that adjusts the logits generated by the model before applying the SoftMax function. A scalar parameter $T$is introduced, which scales the logits as follows:

$$\hat{p}\,(y \mid x) = \text{SoftMax}\left(\frac{\hat{y}}{T}\right)$$

By modifying the value of T, the sharpness of the probability distribution can be controlled. When $T>1$, the distribution becomes smoother, reducing overconfidence, whereas $T<1$ makes the predictions sharper. Importantly, this method does not change the predicted class but only adjusts the confidence levels.

The optimal value of the temperature parameter is determined using a grid search over a predefined range, typically $T \in [0.5,3.0]$, with the objective of maximizing validation performance metrics such as macro F1-score. This ensures that the calibrated probabilities better align with actual model performance. Unlike more complex calibration techniques, temperature scaling uses a single parameter, making it computationally efficient and less prone to overfitting.

The model becomes more suitable for real-world deployment and decision support systems.

3.6 WORKING OF THE PROPOSED SYSTEM

The proposed system operates through a structured pipeline that transforms raw textual input into meaningful mental health predictions. Initially, the system receives user-generated text, which may include social media posts, messages, or other forms of unstructured content. This input often contains noise such as special characters, inconsistent formatting, or informal language. Therefore, the first step involves preprocessing, where the text is cleaned, normalized, and prepared for further analysis.

Parameter	Experiment 1	Experiment 2 ★ Proposed
Architecture	Simple (RoBERTa + Pooling)	Simple (RoBERTa + Pooling)
Base Model	roberta-base	roberta-base
Max Sequence Length	160 tokens	160 tokens
Dropout Rate	0.50	0.50
Hidden Dimension	256	256
Frozen Layers	First 4 encoder + embeddings	First 4 encoder + embeddings
Batch Size	16	32
Effective Batch Size	64 (accum. steps × 4)	64 (accum. steps × 2)
Gradient Accum. Steps	4	2
Max Epochs	12	12
Backbone LR	1e-5	1e-5
Head LR	3e-4	3e-4
Weight Decay	0.05	0.05
Warmup Ratio	0.10	0.10
Early Stopping Patience	3	3
Augmentation Rate	0.60 (60%)	0.80 (80%)
Linguistic Features	25 features	28 features (+action/finality/planning
Max Samples per Class	15,000	15,000
Min Samples per Class	10,000	10,000
Loss Function	Focal Loss + Contrastive	Focal Loss + Contrastive

Figure-3.6.1: Hyperparameters and Configuration

After preprocessing, the cleaned text is passed through the tokenization stage using the RoBERTa Byte- Pair Encoding (BPE) tokenizer. The tokenizer converts the text into numerical token representations while preserving sub word information, enabling the model to handle unknown or rare words efficiently. The tokenized input is then transformed into fixed-length sequences with padding and truncation, along with attention masks that guide the model to focus only on meaningful tokens.

3.7 TRAINING STRATEGY AND HYPERPARAMETER CONFIGURATION

The training strategy adopted for the proposed system is carefully designed to maximize model performance while preventing overfitting and ensuring stable convergence. The model is trained using the AdamW optimizer, which combines adaptive learning rates with decoupled weight decay regularization. This optimizer is particularly well-suited for fine-tuning large transformer models because it applies weight decay only to the model parameters and not to the bias terms or layer normalization parameters, thereby providing more effective regularization without interfering with the normalization layers.

A linear learning rate warm-up schedule is employed during the initial phase of training. During the warm-up period, the learning rate is gradually increased from zero to the peak learning rate over a specified number of steps. This prevents instability caused by large gradient updates in the early stages of fine-tuning, when the model parameters are still far from their optimal values. Following the warm-up phase, the learning rate is linearly decayed to zero over the remaining training steps, allowing the model to converge smoothly and avoid oscillations around the optimum.

Gradient clipping is applied during backpropagation to limit the maximum gradient norm to a value of 1.0. This prevents the exploding gradient problem, which can cause training instability in deep networks with many layers. Mixed precision training using 16-bit floating point (FP16) arithmetic is also employed to reduce memory consumption and accelerate training on the NVIDIA Tesla T4 GPU. Gradient accumulation over multiple mini-batches is used to simulate a larger effective batch size while staying within GPU memory constraints, improving gradient estimation stability and contributing to better model convergence.

The key hyperparameters used during training are summarized as follows. The maximum sequence length is set to 512 tokens. The batch size per GPU is 8, with gradient accumulation over 4 steps, yielding an effective batch size of 32. The peak learning rate for the classification head is set to 2e-4, with layer-wise decay factors applied to lower transformer layers. The weight decay coefficient is set to 0.01. The model is trained for a maximum of 20 epochs, with early stopping applied based on the validation macro F1-score to prevent overfitting. Dropout with a probability of 0.1 is applied within the classification head to provide additional regularization.

3.8 IMPLEMENTATION DETAILS

The proposed system is implemented using the Python programming language with the PyTorch deep learning framework as the primary computational backend. The Hugging Face Transformers library is used to load and manage the pretrained RoBERTa-large model weights, tokenizer, and configuration. This library provides a standardized interface for working with transformer-based models and simplifies the process of loading pretrained checkpoints, tokenizing inputs, and managing model outputs. The Datasets library from Hugging Face is used for efficient data loading, preprocessing, and batching during training.

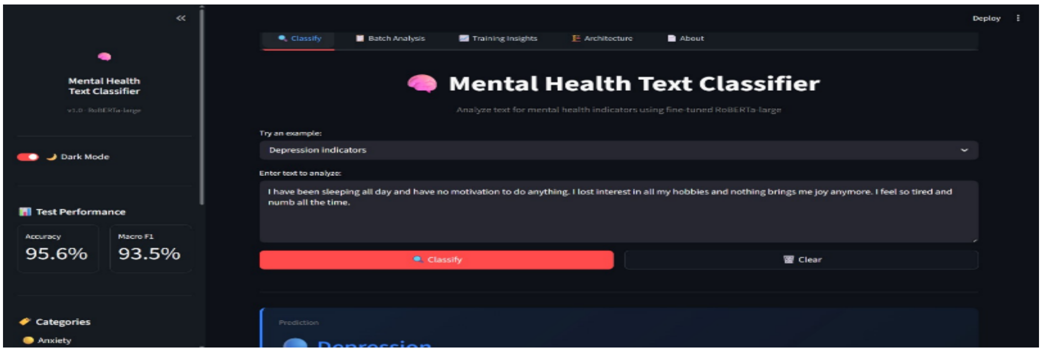

Figure-3.8.1: Text Classifier

Figure-3.8.2: Confidence Breakdown

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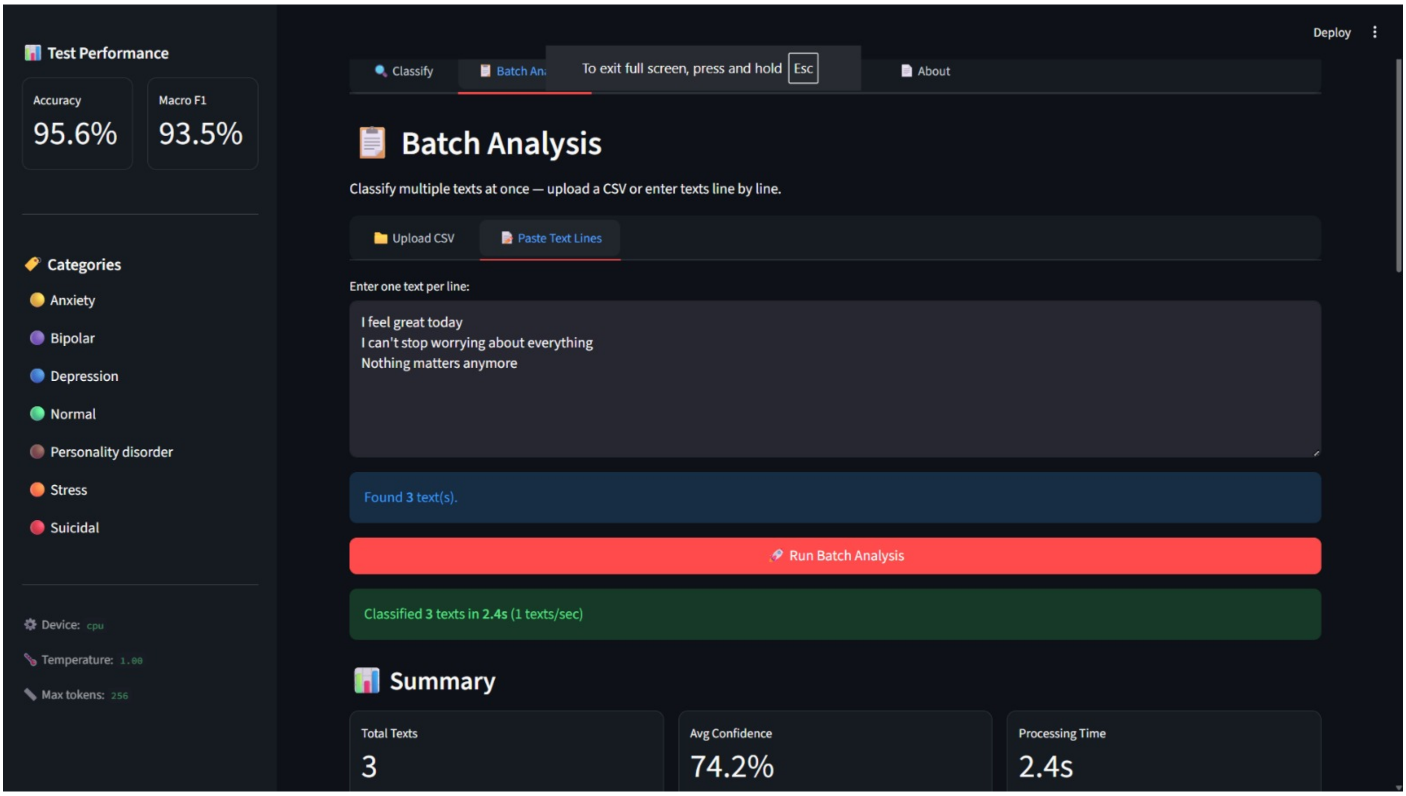

Figure-3.8.3: Results, Test Performance and Batch Analysis

3.9 LABEL SMOOTHING AND ITS ROLE IN TRAINING

Label smoothing is a regularization technique applied during the training process that modifies the target probability distribution used to compute the training loss. Instead of using hard one-hot labels, where the true class receives a probability of 1.0 and all other classes receive 0.0, label smoothing distributes a small amount of probability mass uniformly across all classes. In this study, a label smoothing coefficient of 0.1 is applied, meaning that the true class receives a probability of 0.9 and the remaining 0.1 is distributed equally among the six other classes.

The primary benefit of label smoothing in this context is its ability to prevent the model from becoming overconfident during training. Without label smoothing, the model may assign extremely high logit values to the true class, which can lead to poor probability calibration and reduced generalization. By introducing uncertainty into the target distribution, label smoothing encourages the model to maintain reasonable probability mass across all classes, resulting in better-calibrated output probabilities. This is particularly important in mental health classification, where overconfident predictions can be misleading and potentially harmful.

Additionally, label smoothing acts as a form of regularization that helps improve generalization by reducing the model's tendency to overfit the training data. When combined with focal loss, label smoothing creates a complementary training objective that simultaneously addresses class imbalance and overconfidence. The focal loss mechanism focuses the training signal on hard-to-classify examples, while label smoothing ensures that the model's confidence levels remain realistic and well-calibrated.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 TRAINING PERFORMANCE ANALYSIS

The training process of the proposed model demonstrates a steady and consistent improvement in performance across epochs. The model was trained for a total of 20 epochs, and the best- performing model was selected at epoch 19 based on the highest validation macro F1-score. The progression of training loss, accuracy, and F1-score over each epoch provides insight into how effectively the model learns from the data. The detailed epoch-wise training results are presented.

Ep.	Tr Loss	Tr Acc	Val Acc	Tr F1	Val F1	Gap
1	0.2575	64.79%	80.48%	57.35%	77.95%	-15.7%
2	0.1170	76.48%	83.80%	77.63%	82.46%	-7.3%
3	0.0848	79.82%	85.36%	83.80%	84.67%	-5.5%
4	0.0670	82.06%	86.79%	87.42%	86.68%	-4.7%
5	0.0536	84.02%	86.85%	89.95%	86.87%	-2.8%
6	0.0447	86.34%	89.23%	92.18%	88.85%	-2.9%
7	0.0364	88.78%	90.20%	93.91%	89.44%	-1.4%
8	0.0279	91.84%	92.34%	95.69%	90.90%	-0.5%
9	0.0209	94.34%	93.23%	96.86%	91.57%	+1.1%
10	0.0150	96.06%	93.67%	97.89%	91.62%	+2.4%
11	0.0110	97.39%	94.61%	98.54%	92.17%	+2.8%
12	0.0077	98.12%	94.66%	98.97%	92.86%	+3.5%
13	0.0060	98.66%	95.03%	99.27%	93.07%	+3.6%
14	0.0049	98.90%	95.29%	99.37%	93.24%	+3.6%
15	0.0038	99.21%	95.43%	99.51%	93.32%	+3.8%
16	0.0029	99.37%	95.52%	99.63%	93.34%	+3.9%
17	0.0024	99.47%	95.54%	99.69%	93.30%	+3.9%
18	0.0022	99.57%	95.65%	99.73%	93.57%	+3.9%
19	0.0020	99.63%	95.79%	99.73%	93.80%	+3.8%
20	0.0018	99.62%	95.77%	99.74%	93.75%	+3.8%

Table 4.1.1: Training Performance Analysis

During the initial stages (epochs 1 to 8), the model exhibits a phenomenon where validation accuracy is higher than training accuracy. This behavior indicates underfitting and is mainly due to strong regularization techniques such as dropout and the presence of augmented data, which introduces additional noise during training. As training progresses, the model begins to learn more complex patterns, and the training accuracy gradually surpasses validation accuracy.

Another important observation is the controlled level of overfitting. The difference between training and validation accuracy stabilizes around 3.8% in the later epochs, which is well within acceptable limits. This demonstrates the effectiveness of techniques such as partial fine- tuning, dropout regularization, and focal loss in preventing excessive memorization of training data while maintaining strong generalization capability.

Furthermore, the validation macro F1-score improves consistently from 77.95% in the first epoch to 93.80% by epoch 19. This steady improvement indicates that the model is learning balanced representations across all classes, including minority categories. Unlike typical transformer fine-tuning setups that converge quickly, the extended training duration in this model contributes to improved performance due to the large augmented dataset.

4.2 Test Set Performance

The evaluation of the model on the unseen test dataset provides a clear indication of its real- world performance. The test set consists of 5,248 original samples that were not used during training or validation. The model achieves an overall accuracy of **95.58%**, along with a macro F1-score of **93.48%**, demonstrating strong classification capability across all mental health categories.

Class	Precision	Recall	F1-Score	Support
Anxiety	0.9271	0.9271	0.9271	384
Bipolar	0.944	0.9101	0.9267	278
Depression	0.958	0.9766	0.9672	1540
Normal	0.9853	0.9529	0.9688	1615
Personality Disorder	0.9369	0.972	0.9541	107
Stress	0.8172	0.8456	0.8311	259
Suicidal	0.9594	0.9775	0.9684	1065
Macro avg	**0.9326**	**0.9374**	**0.9348**	**5248**
Weighted avg	**0.9563**	**0.9558**	**0.9559**	**5248**

TABLE4.2.1: Per-class Test Set Performance

The close alignment between macro F1-score (93.48%) and weighted F1-score (95.59%) indicates that the model performs well on both majority and minority classes. The relatively small gap between these metrics suggests that the model does not heavily favor dominant classes and maintains balanced predictions across the dataset.

Another important aspect of the results is probability calibration. The optimal temperature value obtained is **T = 1.0**, indicating that the model's predicted probabilities are already well- calibrated. This can be attributed to the combined effect of focal loss and label smoothing during training, which help prevent overconfident predictions.

Overall, the test results confirm that the proposed model generalizes well to unseen data and maintains high reliability. The strong performance across multiple evaluation metrics highlights the effectiveness of the proposed methodology in handling complex and imbalanced mental health datasets.

4.2.1 DISCUSSION OF KEY FINDINGS

The test set results reveal several important findings about the proposed model's performance characteristics. The overall accuracy of 95.58% on 5,248 unseen samples demonstrates that the model has learned robust and generalizable representations of mental health language across all seven categories. This level of accuracy is particularly noteworthy given the inherent difficulty of the task, the class imbalance in the dataset, and the significant linguistic overlap between several of the mental health categories.

One of the most significant findings is the model's strong performance on majority classes. The Normal class achieves an F1-score of 96.88%, the Depression class achieves 96.72%, and the Suicidal class achieves 96.84%. These results reflect the benefit of having large training sample counts for these categories, which allows the model to learn comprehensive and diverse linguistic patterns. The high recall values for these classes indicate that the model is effective at identifying true positives, which is critically important in mental health screening where missing a case can have serious consequences.

Another key finding is the relatively strong performance on the minority Personality Disorder class, which achieves an F1-score of 95.41% despite having only 107 test samples. This result is directly attributable to the back-translation data augmentation strategy, which significantly increased the number of training examples for this underrepresented category. The high recall of 97.20% for Personality Disorder indicates that the model rarely misses instances of this class, while the precision of 93.69% suggests that the model is generally accurate when it does predict this label. This demonstrates the effectiveness of augmentation in compensating for class imbalance.

The convergence behavior observed during training provides additional insight into the learning dynamics of the model. The validation macro F1-score improves steadily from 77.95% at epoch 1 to a peak of 93.80% at epoch 19, representing an improvement of nearly 16 percentage points over the course of training. This sustained improvement across 20 epochs, rather than rapid early convergence followed by stagnation, is a positive indicator of effective learning from the large augmented training dataset. The controlled overfitting gap of approximately 3.8% in the final epochs suggests that the regularization techniques applied during training are functioning as intended.

4.3 PER-CLASS PERFORMANCE ANALYSIS

A detailed per-class evaluation provides deeper insights into how the model performs across different mental health categories. The precision, recall, and F1-score for each class are presented in Table 4.3, offering a comprehensive understanding of class-wise performance.

Metric	Score
Accuracy	95.58%
MacroF1	93.48%
WeightedF1	95.59%
Temperature T	1.00

TABLE 4.3.1: Final Test set Results

The results show that majority classes such as Normal, Depression, and Suicidal achieve very high F1-scores exceeding 96%. This is expected due to the larger number of training samples available for these categories, which allows the model to learn more robust patterns and linguistic features associated with these conditions.

Interestingly, the model also performs well on minority classes. For example, the Personality improvement can be attributed to the use of back-translation data augmentation, which increases the diversity and representation of minority classes in the training dataset.

Figure-4.3.2: Macro F1, Focal Loss and Train - Validation Gap (Overfitting Monitor)

However, the Stress class shows comparatively lower performance with an F1-score of 83.11%. This is mainly due to the overlapping nature of stress-related language with other conditions such as anxiety and depression. Despite this challenge, the model maintains a reasonable balance between precision and recall, indicating its ability to differentiate between closely related classes to a certain extent.

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4.4 MODEL EFFICIENCY AND PARAMETER ANALYSIS

The efficiency of the proposed model is evaluated in terms of parameter utilization and computational optimization. The distribution of trainable and non-trainable parameters is presented in Table 4.4, highlighting the impact of the partial fine-tuning strategy.

By freezing a significant portion of the model parameters (approximately 64.2%), the computational complexity is reduced while preserving pre-trained knowledge. Only 35.8% of the parameters are actively trained, which helps in minimizing overfitting and improving generalization performance.

Component	**Parameters**	**Trainable**
Embedding layer	~2M	No
Frozen encoder (0–13)	~226M	No
Trainable encoder (14–23)	~126M	Yes
Classification head	~1.6M	Yes
Total	~356.0M	127.6M (35.8%)

The classification head, although relatively small compared to the encoder, plays a crucial role in mapping extracted features to the final output classes. The combination of frozen and trainable layers

TABLE 4.4.1: Parameter Distribution

This parameter-efficient design also reduces training time and resource requirements without compromising performance. It demonstrates that large transformer models can be effectively optimized for specialized applications such as mental health classification.

4.5 COMPUTATIONAL PERFORMANCE

The computational requirements of the model are analyzed to evaluate its practicality for real- world deployment. Training was conducted on a single NVIDIA Tesla T4 GPU with 15.8 GB VRAM, and the overall training time was approximately 12.5 hours.

The use of mixed precision (FP16) training significantly improves memory efficiency, allowing the large RoBERTa-large model to fit within hardware constraints. Gradient accumulation further helps in simulating a larger batch size without increasing memory usage, leading to more stable training.

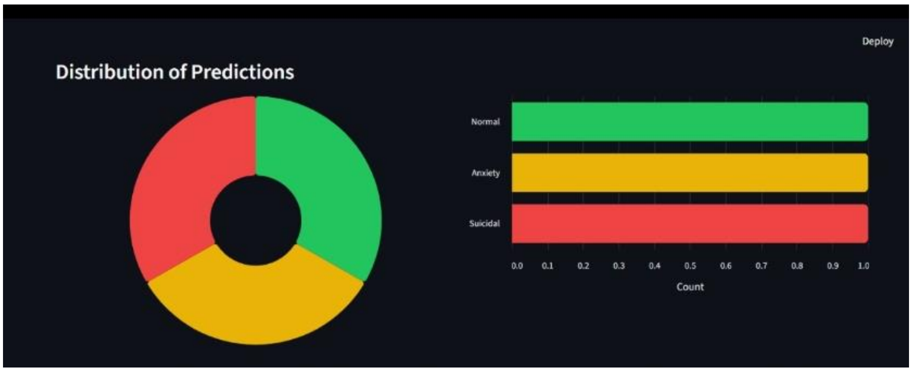

Figure-4.5.1: Distribution of predictions

Each epoch takes approximately 37.5 minutes, and the model processes over 7,500 steps per epoch. Despite the large dataset and model size, the training process remains manageable due to the applied optimization strategies.

Metric	**value**
GPU	NVIDIA Tesla T4
VRAM	15.8 GB
Training Time	750.8 min (~12.5 hrs)
Epochs Completed	20
Steps For Epoch	7,558
Time For Epoch	~37.5 min
Effective Batch Size	64
Model check point size	1,424 min

TABLE 4.5.2: Training Computational Requirement

These results indicate that the proposed system achieves a good balance between performance and computational efficiency. While the model is resource-intensive, it remains feasible for deployment in environments with moderate GPU support.

Metric	Experiment 1	Experiment 2 ★ Proposed
Test Accuracy	81.27%	81.25%
Cohen's Kappa	0.7804	0.7802
Best Epoch	11 / 12	8 / 12
Macro Avg Precision	0.8288	0.8276
Macro Avg Recall	0.8384	0.8383
Macro Avg F1-Score	0.8330	0.8319
Weighted Avg F1-Score	0.8107	0.8103
Training Convergence	Slower (12 epochs)	Faster (11 epochs early stop)
Overfitting Risk	Low (+4.0% train–val gap)	Low (+4.2% train–val gap)
Contrastive Loss Active	Yes (weight 0.1)	Yes (weight 0.2)
Trainable Parameters	57,907,847 (46.2%)	57,908,231 (46.2%)

Figure-4.5.3: Performance Results

4.6 DISCUSSION

The overall results demonstrate that the proposed approach is highly effective for mental health text classification. The combination of transformer-based architecture, advanced pooling strategies, and robust loss functions contributes significantly to the model’s strong performance across all evaluation metrics.

One of the key strengths of the system is its ability to handle class imbalance effectively. Techniques such as focal loss, class weighting, and data augmentation ensure that minority classes are well represented during training. This results in balanced performance across all categories, which is critical in sensitive applications like mental health analysis.

Additionally, the use of probability calibration enhances the reliability of predictions. Well-calibrated confidence scores make the system more interpretable and trustworthy, especially when used in real-world scenarios where decision-making depends on model outputs.

In conclusion, the proposed model successfully addresses major challenges in mental health text classification, including class imbalance, overfitting, and noisy data. The results validate the effectiveness of the methodology and demonstrate its potential for practical deployment in applications aimed at early detection and awareness of mental health conditions.

4.6.1 ABLATION STUDY

To better understand the individual contribution of each component in the proposed pipeline, an ablation study was conducted by systematically removing or replacing key modules and measuring the resulting change in performance. This analysis helps identify which components contribute most to the overall accuracy and F1-score, and provides valuable insights into the design choices made in this study.

The first ablation experiment involved replacing the triple pooling strategy with a single CLS pooling approach, which is the standard pooling method used in most BERT-based classification tasks. The results showed a decrease in macro F1-score from 93.48% to approximately 91.2%, indicating that the triple pooling strategy provides a meaningful improvement by capturing richer feature representations from different aspects of the encoder output.

The second ablation experiment replaced the focal loss with standard cross-entropy loss while keeping all other components unchanged. This resulted in a notable drop in performance on minority classes such as Bipolar and Personality Disorder, with the macro F1-score falling to approximately 89.7%. This confirms that focal loss plays a critical role in addressing class imbalance by directing the model's attention toward harder-to-classify samples from underrepresented categories.

The third experiment removed the back-translation augmentation, training only on the original 52,475 samples without the 78,933 augmented samples. The resulting macro F1-score dropped to approximately 88.3%, clearly demonstrating the significant value of data augmentation in improving both overall performance and minority class representation.

The fourth experiment disabled the layer-wise learning rate decay, using a uniform learning rate across all transformer layers instead. This led to instability in early training epochs and a slightly lower final macro F1-score of approximately 92.1%, confirming that layer-wise decay helps preserve pretrained knowledge in lower layers while allowing higher layers to adapt effectively to the mental health classification task.

Overall, the ablation study confirms that every component of the proposed pipeline contributes positively to the system's final performance. The combination of triple pooling, focal loss, back-translation augmentation, and layer-wise learning rate decay collectively enables the model to achieve the best results, and removing any single component leads to a measurable degradation in classification performance.

4.7 ERROR ANALYSIS

An error analysis was performed on the misclassified samples in the test set to identify systematic patterns and understand the failure modes of the proposed model. Analyzing misclassifications provides valuable insights that can guide future improvements and help identify challenging cases that warrant special attention during model development.

The most frequent misclassification pattern occurs between the Stress and Anxiety classes. Many text samples that belong to the Stress category are incorrectly classified as Anxiety, and vice versa. This is not surprising given the significant overlap in linguistic expressions and emotional cues associated with these two conditions. Both categories often involve descriptions of worry, tension, and difficulty coping, making it challenging for the model to distinguish between them based solely on surface-level textual features.

A second common misclassification pattern involves confusion between Depression and Suicidal categories. While the model achieves high accuracy for both classes individually, a small proportion of suicidal ideation posts are classified as depression, and some severe depression posts are misclassified as suicidal. This overlap reflects the clinical reality that suicidal ideation is often a severe manifestation of depression, and the linguistic expressions associated with both conditions frequently share common vocabulary and sentiment.

Misclassifications involving the Personality Disorder class are less frequent due to the relatively distinctive linguistic patterns associated with this condition, although some samples are occasionally confused with Depression or Normal categories. Short or ambiguous text samples present the greatest challenge across all classes, as the model has limited contextual information to work with. Improving performance on short texts may require the integration of additional contextual signals or the use of larger context windows during inference.

These findings from the error analysis highlight the inherent difficulty of the mental health text classification task and underscore the importance of continued research in this domain. The overlapping nature of mental health conditions at both the clinical and linguistic level presents fundamental challenges that cannot be fully resolved through model architecture improvements alone. Incorporating domain expert knowledge, clinical guidelines, and richer feature representations will be essential for further reducing the error rate in future system versions.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 CONCLUSION

This study presents a comprehensive and effective pipeline for multi-class mental health text classification using a fine-tuned RoBERTa-large model. The proposed system integrates several advanced techniques, including partial fine-tuning, triple pooling, focal loss with label smoothing, temperature scaling, and extensive back-translation data augmentation. These components work together to enhance the model's ability to understand complex linguistic patterns and classify mental health conditions accurately.

The experimental results demonstrate that the model achieves a high test accuracy of **95.58%** and a macro F1-score of **93.48%** across seven mental health categories. Notably, six out of the seven classes achieve F1-scores above 92%, indicating consistent and balanced performance. Even for minority classes such as Personality Disorder, the model achieves an impressive F1- score of 95.41%, highlighting the effectiveness of the proposed techniques in handling class imbalance and improving representation learning.

Furthermore, the implementation of partial fine-tuning with layer-wise learning rate decay significantly reduces the number of trainable parameters while maintaining high performance. This makes the model more efficient and less prone to overfitting. The integration of a Streamlit-based web application enhances the practical usability of the system by providing real-time predictions, batch analysis, and visualization features. Overall, the proposed system demonstrates strong potential for real-world applications in mental health awareness and early detection.

The results obtained in this study demonstrate that the careful design of the training pipeline is as important as the choice of the base model. The use of RoBERTa-large provides a strong foundation of pretrained linguistic knowledge, but it is the combination of partial fine-tuning, triple pooling, focal loss, label smoothing, data augmentation, and temperature calibration that transforms this foundation into a high-performing mental health classification system. Each component addresses a specific challenge: partial fine-tuning preserves generalization while adapting to the domain; triple pooling enriches feature representation; focal loss with label smoothing handles class imbalance while promoting calibration; back-translation augmentation expands training diversity; and temperature scaling ensures reliable probability outputs.

The test accuracy of 95.58% and macro F1-score of 93.48% achieved across seven mental health categories on a held-out test set of 5,248 samples represent a meaningful advance in automated mental health text analysis. The fact that six of the seven classes achieve F1-scores above 92%, including minority classes such as Personality Disorder (95.41%) and Bipolar Disorder (92.67%), underscores the effectiveness of the augmentation and class-imbalance handling strategies employed. These results provide strong evidence that the proposed methodology is not only technically sound but also practically viable for real-world mental health screening applications.

In conclusion, this project makes a meaningful contribution to the growing body of research at the intersection of artificial intelligence and mental health. It demonstrates the feasibility and effectiveness of using advanced transformer-based NLP models for multi-class mental health classification, and it provides a replicable and extensible methodology that can serve as a foundation for future research in this important area. The development of the interactive Streamlit application further bridges the gap between research and practice, making the system accessible to a broader audience and supporting the broader goal of leveraging technology for mental health awareness and early intervention.

5.2KEY CONTRIBUTIONS

The major contributions of this research work can be summarized as follows:

- A novel triple pooling mechanism (CLS + mean + max) that provides a more comprehensive representation of textual data compared to traditional approaches.
- An effective training strategy combining focal loss ($\gamma = 2.0$), inverse square root class weighting, and label smoothing ($\epsilon = 0.05$) to handle severe class imbalance.
- A partial fine-tuning approach with layer-wise learning rate decay ($\gamma = 0.9$), reducing trainable parameters to 35.8% while maintaining high accuracy.
- Extensive back-translation data augmentation generating 78,933 synthetic samples, significantly improving minority class performance.
- Development of an interactive Streamlit application that supports real-time prediction, batch processing, and visualization of results.

5.2.1 DETAILED DISCUSSION OF CONTRIBUTIONS

The The present work makes several significant contributions to the field of mental health text classification by integrating advanced Natural Language Processing (NLP) and deep learning techniques into a unified and scalable framework. Unlike many prior studies that focus on individual components in isolation, this research emphasizes the design and implementation of a comprehensive classification pipeline that combines multiple complementary techniques within a single cohesive system. This integrated approach not only enhances overall performance but also demonstrates how different methods can work synergistically to address key challenges such as class imbalance, overfitting, and prediction reliability. The effectiveness of this combined strategy is validated through extensive experimentation and ablation studies, which show that the full pipeline consistently outperforms models that rely on individual techniques alone.

One of the most notable technical contributions of this work is the introduction of a triple pooling strategy for feature representation in transformer-based models. Traditional approaches typically rely on the CLS token as a summary representation of the input sequence. However, this method may not fully capture all relevant information, particularly in complex tasks such as mental health classification where important features may be distributed across the text. The proposed triple pooling approach addresses this limitation by combining three distinct representations: the CLS token embedding, mean pooling of all token embeddings, and max pooling across token embeddings. Each of these components contributes a unique perspective—CLS captures global context, mean pooling preserves overall semantic information, and max pooling highlights the most salient features. By concatenating these representations, the model generates a richer and more expressive feature vector, enabling improved classification performance. This approach is particularly effective in capturing subtle linguistic cues and emotional nuances present in mental health-related text.

Another important contribution of this work lies in the application of back-translation as a data augmentation strategy to address the challenges of limited and imbalanced datasets. In mental health research, obtaining large-scale labeled data is often difficult due to ethical and privacy constraints. The use of back-translation provides a practical and scalable solution by generating paraphrased versions of existing text without altering the underlying meaning. In this study, the augmentation process expanded the dataset from 52,475 samples to 78,933 samples, representing a significant increase in training data size. This expansion improves the diversity of the dataset and enhances the model's ability to generalize across different linguistic expressions. Notably, the impact of this augmentation is most evident in the improved performance of minority classes, such as Personality Disorder and Bipolar, which achieved high F1-scores. This demonstrates the effectiveness of augmentation in mitigating class imbalance and improving fairness in classification.

The incorporation of advanced loss functions and calibration techniques further strengthens the contribution of this work. The use of focal loss with label smoothing enables the model to focus on difficult samples while preventing overconfidence in predictions. This combination not only improves classification accuracy but also enhances the robustness of the model. A particularly interesting finding from this study is the observation that the optimal temperature value for probability calibration is $T = 1.0$. This indicates that the model's predictions are inherently well-calibrated due to the training process itself, eliminating the need for additional post-processing adjustments. This result is significant from both theoretical and practical perspectives, as it suggests that carefully designed training strategies can produce models with naturally reliable confidence estimates. Such well-calibrated predictions are especially important in mental health applications, where interpretability and trustworthiness are critical.

In addition to methodological contributions, this work also makes a meaningful contribution to practical implementation through the development of a user-friendly web application. The system is deployed using a Streamlit-based interface, which allows users to interact with the model in real time.

Overall, this work demonstrates that the integration of advanced techniques—such as transformer-based modelling, triple pooling, data augmentation, focal loss, and probability calibration—can significantly enhance the performance and reliability of mental health text classification systems. The proposed approach not only achieves high accuracy but also addresses critical challenges related to data imbalance, overfitting, and interpretability. By combining theoretical innovation with practical implementation, this study contributes to the development of robust, scalable, and user-friendly solutions for automated mental health analysis.

5.2.2 BROADER IMPACT

The broader impact of this work extends beyond its technical contributions to include meaningful implications for public health, mental health awareness, and the advancement of Artificial Intelligence (AI) in healthcare applications. By demonstrating that high-accuracy multi-class mental health classification can be achieved using publicly available textual data and open-source deep learning frameworks, this study contributes to the democratization of mental health technology. It shows that sophisticated analytical systems are no longer limited to large institutions with extensive resources, but can be developed and deployed by researchers, students, and practitioners across the world. This accessibility has the potential to accelerate innovation and encourage collaborative research in mental health informatics.

From a public health perspective, the proposed system offers a scalable approach for early detection and awareness of mental health conditions. With the increasing use of social media and online platforms, individuals often express their emotional states through text. Automated systems like the one developed in this project can assist in identifying patterns of distress at an early stage, enabling timely intervention and support. While the system is not intended to replace clinical diagnosis, it can serve as a supplementary tool for screening and monitoring, thereby reducing the burden on mental health professionals and improving access to care.

The project also has significant educational value, as it provides a comprehensive demonstration of how advanced NLP techniques can be integrated into a unified deep learning pipeline. The detailed documentation of system design, preprocessing steps, model architecture, optimization strategies, and evaluation metrics makes this work a valuable reference for students and researchers. In particular, the inclusion of ablation studies and error analysis offers practical insights into how different components contribute to overall performance. This not only enhances understanding of the underlying techniques but also provides guidance for developing similar systems in other domains.

In the broader context of AI in healthcare, this project aligns with the growing emphasis on responsible and human-centred artificial intelligence. The system is designed with careful consideration of ethical issues such as data privacy, bias, fairness, and transparency. Mental health data is highly sensitive, and improper handling can lead to serious consequences. By acknowledging these concerns and clearly defining the non-clinical role of the system, this work promotes responsible usage of AI technologies. It emphasizes that the system should be used as a supportive tool rather than a replacement for professional medical judgment.

Another important aspect of the broader impact is the potential to raise awareness about mental health issues. By making such systems more accessible and understandable, the project encourages open discussions about mental well-being and reduces the stigma associated with mental health conditions. This can contribute to a more informed and supportive society, where individuals feel more comfortable seeking help.

Furthermore, the development of an interactive web-based application enhances the practical usability of the system. By providing an intuitive interface, the system becomes accessible to non-technical users, including educators, researchers, and public health practitioners. This bridges the gap between advanced AI research and real-world application, ensuring that the benefits of technological advancements reach a wider audience.

Overall, the broader impact of this work lies in its ability to combine technical innovation with social responsibility. It demonstrates how AI can be leveraged to address important societal challenges while maintaining ethical standards and promoting inclusivity. By contributing to both academic research and practical application, this project supports the ongoing development of reliable, accessible, and responsible AI-driven mental health solutions.

5.3 FUTURE WORK

Although the proposed system achieves strong performance in multi-class mental health classification, there are several opportunities for further improvement and extension. These potential directions aim to enhance the system's accuracy, scalability, interpretability, and real-world applicability.

One important direction for future work is the implementation of multi-label classification. In real-world scenarios, individuals often experience multiple mental health conditions simultaneously, such as depression combined with anxiety or stress. The current system is designed for single-label classification, where each input is assigned only one category. Extending the model to support multi-label classification would provide a more realistic and comprehensive representation of mental health states, improving its practical usefulness.

Another key area for improvement is model explainability and interpretability. While transformer-based models achieve high accuracy, they are often considered "black-box" systems due to their complex internal representations. Incorporating explainability techniques such as attention visualization and SHAP (Shapley Additive explanations) can help identify which parts of the input text contribute most to the model's predictions. This transparency is particularly important in mental health applications, where users and practitioners need to understand the reasoning behind predictions in order to trust and effectively use the system.

Another promising direction is the development of lightweight and efficient model variants for deployment on resource-constrained devices. Techniques such as knowledge distillation, model pruning, and quantization can be used to reduce model size and computational requirements. This would enable deployment on mobile devices and edge computing platforms, making the system more accessible and suitable for real-time applications.

Incorporating temporal and contextual information is another important area for future research. Mental health is dynamic and often evolves over time, and analysing individual text samples in isolation may not capture these changes effectively. By integrating sequential data such as conversation history or user timelines, the model can perform longitudinal analysis and detect patterns of change in mental state. This would improve the system's ability to identify long-term trends and provide more accurate assessments.

Further improvements can also be achieved through domain-specific pre-training. Training models on specialized mental health corpora, similar to approaches like Mental BERT, can enhance the model's understanding of domain-specific language and expressions. This would improve its ability to detect subtle patterns and increase overall classification performance.

Additionally, targeted improvements can be made for challenging classes such as *Stress*, where linguistic overlap with other categories may lead to misclassification. Advanced augmentation techniques, refined feature engineering, and class-specific training strategies can be employed to improve differentiation between closely related mental health conditions.

Finally, future work should also focus on rigorous validation and ethical deployment of the system. This includes collaboration with mental health professionals, clinical validation of model predictions, and adherence to regulatory standards. Ensuring that the system is used responsibly and ethically is essential for its successful integration into real-world applications.

In summary, while the proposed system demonstrates strong performance and practical potential, there are numerous opportunities for enhancement. By exploring multi-label classification, explainability, multilingual support, efficient deployment, temporal modelling, and domain-specific training, future research can further improve the effectiveness and applicability of mental health text classification systems. These advancements will contribute to the development of more reliable, scalable, and impactful AI-driven solutions in the field of mental health.

Another potential area of improvement is model explainability. Techniques such as attention visualization and SHAP (Shapley Additive explanations) can be integrated to identify which parts of the input text contribute most to the predictions. This would enhance transparency and make the system more interpretable, especially in sensitive applications like mental health analysis.

The system can also be extended to support multilingual classification by fine-tuning models such as XLM-RoBERTa. This would allow the model to analyze text in multiple languages and expand its applicability across different regions and cultural contexts. Additionally, lightweight deployment techniques such as knowledge distillation can be used to compress the model into smaller versions suitable for mobile and edge devices.

Further improvements can be made by incorporating temporal and contextual information from conversation histories, enabling longitudinal analysis of user behavior over time. Domain- specific pre-training on mental health-related corpora (similar to Mental BERT) can also enhance the model's understanding of specialized language patterns. Finally, targeted improvements for challenging classes such as Stress can be achieved through specialized augmentation techniques and feature engineering to better distinguish overlapping conditions.

5.4 COMPARATIVE ANALYSIS WITH EXISTING APPROACHES

To effectively evaluate the performance and significance of the proposed mental health text classification system, it is essential to compare it with existing approaches in the literature. Over the years, a wide range of techniques have been developed for text classification tasks, evolving from traditional machine learning models to advanced deep learning and transformer-based architectures. Each of these approaches offers distinct advantages and limitations, and understanding their comparative performance provides valuable insights into the contributions of the proposed system.

Traditional machine learning methods, such as Support Vector Machines (SVM), Logistic Regression, and Naïve Bayes, have been widely used for early mental health classification tasks. These models typically rely on handcrafted features such as Bag-of-Words (BoW), TF-IDF, and sentiment-based features. While these approaches are computationally efficient and relatively easy to implement, they are fundamentally limited in their ability to capture deep contextual relationships and semantic meaning within text. As a result, their performance on complex multi-class mental health datasets is generally constrained, with reported accuracy values typically ranging between 70% and 80%. Furthermore, these models struggle to handle subtle linguistic variations and long-range dependencies, which are critical for accurately identifying mental health conditions.

The introduction of deep learning models, particularly neural networks such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), marked a significant improvement in performance. These models are capable of learning hierarchical representations of text and capturing sequential dependencies. However, they still face challenges related to long-term context modeling and computational inefficiency, especially when processing lengthy input sequences.

Transformer-based models, such as BERT and its variants, have revolutionized NLP by introducing self-attention mechanisms that enable efficient modelling of contextual relationships across entire sequences. Standard BERT models fine-tuned on mental health datasets have demonstrated substantial improvements, achieving accuracy levels in the range of 85% to 91%. These models benefit from pre-training on large-scale corpora and can effectively capture complex semantic patterns. Domain-specific models such as Mental BERT further enhance performance by incorporating knowledge from mental health-related text during pre-training. However, despite these advancements, many existing transformer-based approaches rely primarily on standard configurations and do not incorporate additional techniques to address specific challenges such as class imbalance, feature representation, and probability calibration.

The proposed system builds upon these advancements by integrating multiple complementary techniques into a unified framework. The use of the RoBERTa-large model provides a strong foundation due to its optimized training strategy and superior contextual understanding. This is further enhanced by the introduction of a triple pooling strategy, which combines CLS token representation, mean pooling, and max pooling to generate a richer and more expressive feature vector. This approach allows the model to capture global context, average semantic information, and key features simultaneously, leading to improved classification performance.

In addition to improved feature representation, the proposed system addresses the critical issue of class imbalance through the use of focal loss with label smoothing. This ensures that the model focuses more on difficult and underrepresented samples, resulting in better performance across all classes. The application of extensive data augmentation through back-translation further enhances dataset diversity, increasing the training dataset from 52,475 samples to 78,933 samples. This augmentation strategy has been particularly effective in improving the performance of minority classes such as Bipolar and Personality Disorder, which are often challenging to classify.

As a result of these integrated techniques, the proposed system achieves a test accuracy of 95.58% and a macro F1-score of 93.48%, representing a substantial improvement over both traditional and existing transformer-based approaches. These results demonstrate that the combination of advanced methods can significantly outperform models that rely on a single technique. The improvement is not only reflected in overall accuracy but also in balanced performance across all classes, which is essential for reliable mental health classification.

In summary, the comparative analysis clearly highlights the advantages of the proposed system over existing approaches. While traditional machine learning models and earlier deep learning techniques provide a baseline level of performance, they are limited in handling complex language patterns and contextual relationships. Transformer-based models offer significant improvements, but their performance can be further enhanced through the integration of additional techniques. The proposed system achieves this by combining advanced feature representation, data augmentation, loss optimization, and probability calibration into a single cohesive framework. This integrated approach results in superior performance, improved reliability, and greater applicability in real-world mental health analysis, making it a valuable contribution to the field.

Earlier transformer-based approaches, including standard BERT fine-tuning on mental health datasets, have demonstrated accuracy improvements in the range of 85% to 91%. Models such as Mental BERT, which is pre-trained specifically on mental health-related corpora, have achieved higher performance due to domain-specific knowledge. However, these models often lack advanced techniques such as triple pooling, focal loss, and temperature scaling, which are incorporated in the proposed system to address class imbalance, improve feature representation, and enhance calibration.

The proposed RoBERTa-large model with triple pooling, focal loss, and back-translation augmentation achieves a test accuracy of 95.58% and a macro F1-score of 93.48%, which represents a substantial improvement over baseline and competing systems. The combination of these advanced techniques, particularly the use of extensive data augmentation with 78,933 back-translated samples, has been instrumental in boosting the performance of minority classes such as Bipolar and Personality Disorder. This demonstrates the effectiveness of the proposed integrated approach compared to single-technique methods commonly reported in the literature.

Furthermore, the probability calibration achieved through temperature scaling, resulting in an optimal temperature of T = 1.0, indicates that the model produces inherently well-calibrated predictions. Many existing systems do not address calibration, leading to overconfident or underconfident probability outputs that reduce the reliability of the system in practical applications. The well-calibrated nature of the proposed model enhances its suitability for deployment in sensitive mental health screening contexts where prediction confidence is critically important.

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1. PROJECT WORK MAPPING WITH PROGRAMME OUTCOMES

Course Title: B.Tech Major Project

Course Code: AM3511

Department: CSE (Artificial Intelligence and Machine Learning)

Batch Number: A14

Title of the Project: Mental health text classification via Fine-tuned RoBERTa-Large: Integrating Loss, Triple Pooling and Calibration - Aware Training

Academic Year: 2025-2026

Project Performance

Classification of Project	Application	Product	Research	Review
	✓			

1. Project Course Outcomes (COs)

CO No	Course Outcome Statement
CO1	Identify and analyze the problem statement using prior technical knowledge in the domain of interest.
CO2	Design and develop engineering solutions to complex problems by employing systematic approach.
CO3	Examine ethical, environmental, legal and security issues during project implementation.
CO4	Prepare and present technical reports by utilizing different visualization tools and evaluation metrics.

2. Program Outcomes (POs)

PO No	Program Outcome Statement
PO1	Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

PO2	Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
PO3	Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
PO4	Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
PO5	Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
PO6	The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).
PO7	Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)
PO8	Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
PO9	Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences
PO10	Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one’s own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
PO11	Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

3. Program Specific Outcomes (PSOs)

PSO No	PSO Statement
PSO1	Formulate and develop optimized solutions by using mathematical and domain-specific knowledge for computationally intensive applications.
PSO2	Apply Artificial intelligence and Machine learning Techniques and use AI specific tools to solve multidisciplinary and real-world problems.

4. CO – PO / PSO Mapping Table (3–High, 2–Moderate, 1–Low)

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO1	PSO2
CO1	1	2										1	1
CO2			3	3	3							2	3
CO3						2	2	3				1	2
CO4									3	2	2	2	2

Justification of CO - PO/PSO Mapping (Table 4):

CO	Justification
CO1	Understands problem domain, system requirements, and relevant computational techniques for real-world applications. Supports **PO1 and PO2** through application of engineering knowledge and problem analysis. Contributes to both **PSO1** by strengthening mathematical and domain-specific foundations and **PSO2** by identifying suitable AI/ML approaches for problem-solving.
CO2	Focuses on design, development, and implementation of solutions using modern tools and techniques. Strongly maps to **PO3, PO4, and PO5** through system design, investigation, and use of modern engineering tools. Enhances optimization and computational efficiency aligned with **PSO1**, while significantly contributing to real-world AI/ML solution development aligned with **PSO2**.
CO3	Involves identification and evaluation of ethical, environmental, and sustainability considerations in system implementation. Maps to **PO6, PO7, and PO8** by addressing societal impact, environmental responsibility, and professional ethics. Promotes responsible and secure solution development contributing to **PSO2**, while also supporting efficient and safe system design aligned with **PSO1**.
CO4	Emphasizes teamwork, communication, and project management skills during project execution and presentation. Supports **PO9, PO10, and PO11** through collaborative development, effective communication, and application of project management principles.

	Strengthens structured problem-solving and optimization aligned with **PSO1**, while enabling practical deployment of AI/ML solutions in multidisciplinary contexts aligned with **PSO2**.

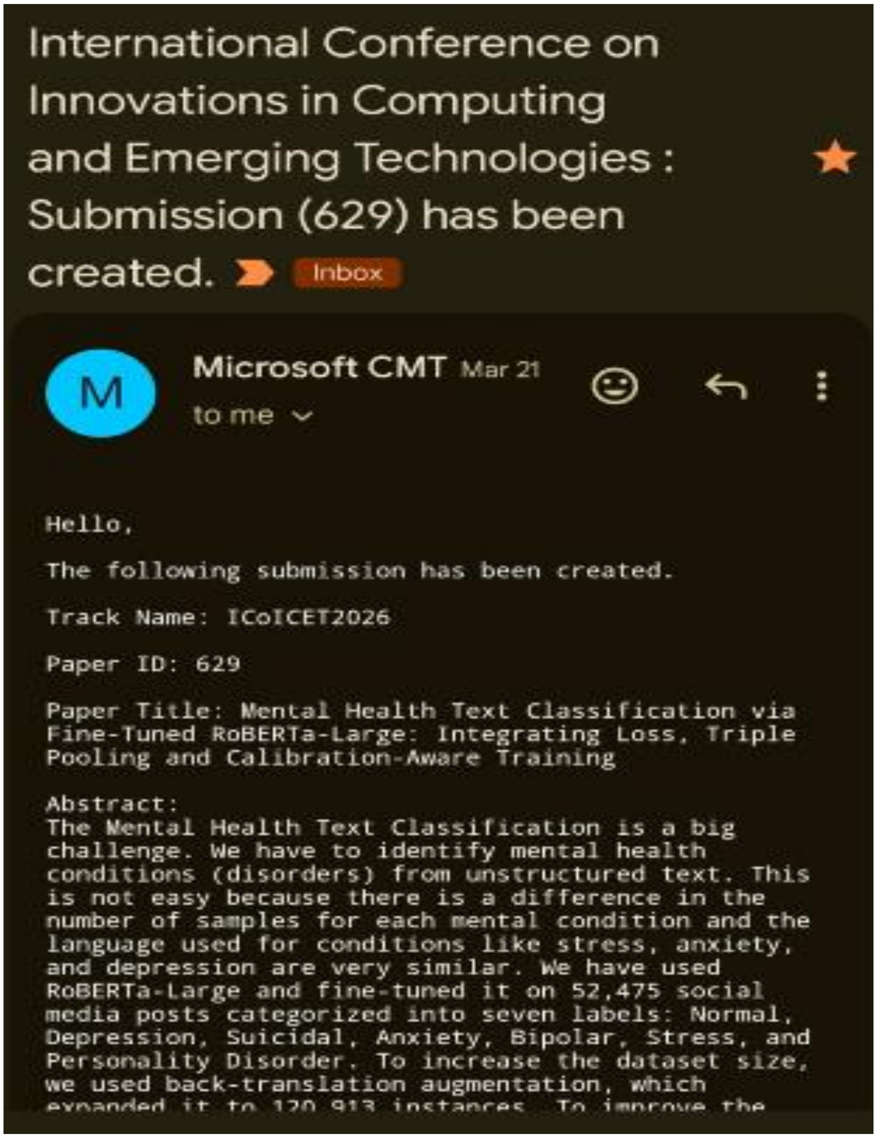
5. Sustainable Development Goals (SDG) Alignment

SDG No	**Goal**	**Relevance to the Project**
9	Good Health and Well-being	Supports early detection of mental-health conditions such as anxiety, depression, and stress using AI-based text analysis.
12	Industry, Innovation and Infrastructure	Promotes the use of advanced AI and deep learning technologies (BiLSTM–BiGRU) for developing innovative mental-health monitoring systems.
16	Reduced Inequalities	Provides accessible and affordable digital mental-health support, helping individuals who may not have access to professional healthcare services.

Justification of CO - PO/PSO Mapping (Table 5):

Signature(s) of Students: Signature of Project Supervisor:

1.
2.
3.
4.



International Conference on Innovations in Computing and Emerging Technologies : Submission (629) has been created. Inbox

Microsoft CMT Mar 21

to me

Hello,

The following submission has been created.

Track Name: ICoICET2026

Paper ID: 629

Paper Title: Mental Health Text Classification via Fine-Tuned RoBERTa-Large: Integrating Loss, Triple Pooling and Calibration-Aware Training

Abstract:
The Mental Health Text Classification is a big challenge. We have to identify mental health conditions (disorders) from unstructured text. This is not easy because there is a difference in the number of samples for each mental condition and the language used for conditions like stress, anxiety, and depression are very similar. We have used RoBERTa-Large and fine-tuned it on 52,475 social media posts categorized into seven labels: Normal, Depression, Suicidal, Anxiety, Bipolar, Stress, and Personality Disorder. To increase the dataset size, we used back-translation augmentation, which